A linear Erdős–Heilbronn inequality over arbitrary fields

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Abstract

Let F be a field, $A \subseteq F$ a finite set of size $m \geq n$, and $a_1, \ldots, a_n$ nonzero weights. We prove that the sums $\sum_i a_i x_i$, with pairwise distinct $x_i \in A$, have cardinality at least $\min\{p(F) - \delta, n(m-n)+1\}$, provided $p(F) \neq n+1$. Here $p(F)$ is the characteristic, interpreted as infinity in characteristic zero, and $\delta = 1$ exactly when $n = 2$ and $a_1 + a_2 = 0$. A minimal-counterexample argument gives an exact reduction to prime fields. Full Taylor insertions into an alternating identity then reduce the remaining problem to capacity frontiers. Uniform cost bounds handle unbounded parameters, and explicit finite integer and polynomial certificates close the bounded domains. All additional algebraic lemmas and exact verification programs are included.

Keywords: restricted sumsets; weighted sums; Erdős–Heilbronn inequality; polynomial method; finite fields.

Part I
Reduction from arbitrary fields to prime fields

I.1 Statement and main theorem

For a field F, write $p(F) = \mathrm{char}(F)$, with $p(F) = \infty$ in characteristic zero. Given a finite set $A \subseteq F$, $m = |A| \geq n \geq 1$, and nonzero weights $a_1, \ldots, a_n$, put

$$S(a; A) = \left\{ \sum_{i=1}^{n} a_i x_i : x_i \in A,\ x_i \neq x_j\ (i \neq j) \right\}.$$

The choices $(x_1, \ldots, x_n)$ are ordered; repetitions are removed only from the resulting set of sums. Let $\mathsf{EH}(F)$ denote the following assertion for all such data: if $p(F) \neq n+1$, then

$$|S(a; A)| \geq \min\{p(F) - \delta, n(m-n)+1\}, \qquad \delta = \begin{cases} 1, & n = 2,\ a_1 + a_2 = 0, \\ 0, & \text{otherwise.} \end{cases} \tag{I.1.1}$$

Here $\infty - \delta = \infty$. No assumptions that F is finite, $A \subseteq \mathbb{F}_p$, $n < p$, or $m < p$ are imposed.

Theorem I.1.1 (Linear Erdős–Heilbronn inequality)**.** *Under the preceding hypotheses, inequality* (I.1.1) *holds.*

The proof is completed in Section IV.1. We first establish a prime-field reduction, and then prove the required assertion over every prime field.

Theorem I.1.2 (Prime-field reduction)**.** *For each prime p, the assertion $\mathsf{EH}(\mathbb{F}_p)$ is equivalent to $\mathsf{EH}(F)$ for every field F of characteristic p. Together with the known characteristic-zero theorem, this reduces the assertion over all fields to its instances over prime fields.*

More precisely, fix a field F admitting a counterexample, and minimize n over all admissible counterexamples in this same field, allowing arbitrary finite sets, weights, and excesses $m-n$. Then F has positive prime characteristic p. Set

$$W = \{a_1, \dots, a_n\}, \quad U = \operatorname{span}_{\mathbb{F}_p}(A - A),$$
$$L = \operatorname{span}_{\mathbb{F}_p}\{(w - w')(x - y) : w, w' \in W,\ x, y \in A\}.$$

Then $\dim_{\mathbb{F}_p} U = \dim_{\mathbb{F}_p} L = 1$, and there exist $\alpha \in F$ and $u, c \in F^*$ such that

$$B = (A - \alpha)/u \subseteq \mathbb{F}_p, \qquad b_i = a_i/c \in \mathbb{F}_p^*, \qquad S(b; B) = \frac{S(a; A) - \alpha \sum_i a_i}{cu}. \tag{I.1.2}$$

Thus $n, m, p, |S|$, and every equality class and multiplicity of weights are preserved exactly.

The assertion concerns a minimal counterexample in a fixed source field. It does not assert this model for arbitrary counterexamples, nor minimality of the normalized model over $\mathbb{F}_p$.

We derive numerical and translation-boundary constraints for a minimal counterexample, and show that its difference-product space has dimension at most two. Compression growth and correlation counts exclude dimension two: 72 exact polynomial identities first bound the parameters, and 3,766 endpoint certificates eliminate the remaining finite domain. A rectangle of genuine increments then gives the exact prime-field model.

I.2 Numerical constraints for a minimal counterexample

Every appeal to minimality below is in the same field. A problem with $n' < n$ variables is covered only after checking its characteristic exclusion and correction term. Minimality ranges over all excesses, not a fixed value of $m - n$.

Lemma I.2.1 (Minimal core)**.** *For a minimal counterexample as in Theorem I.1.2, put*

$$h = m - n, \quad s = |S|, \quad K_0 = (n-1)h + 1, \quad V = \min(nh, p-1), \quad b_h = 2h - 1.$$

Then

$$h \geq 3, \quad n \geq 6, \quad h + 2 \leq n < 2h^2 - h, \quad K_0 \leq s \leq V, \quad K_0 < p, \tag{I.2.1}$$
$$p - m \geq \max\{8, (n-2)(h-1)\}, \tag{I.2.2}$$
$$2p < n(3n-5), \qquad p \leq nh + \binom{n}{2}, \tag{I.2.3}$$
$$h = n - 2 \implies p < (n-1)^2. \tag{I.2.4}$$

Every nonzero weight has multiplicity at most h. In particular, the weights are not all equal, $p \geq 17$, $m < p$, $n \leq p - 2$, and $\delta = 0$.

Appendix A supplies the proof, including large sets, zero padding, the two-variable correction, distinct-weight blocks, and the residual five-variable case. Appendix B proves the six-point polynomial identities required in that residual case.

I.3 Genuine swap boundaries and dimension constraints

For finite $E \subseteq F$ and $t \in F$, define

$$\lambda_E(t) = |E \setminus (E+t)| = |E| - |E \cap (E+t)|.$$

Then $\lambda_E(-t) = \lambda_E(t)$ and

$$\lambda_E(t+v) \le \lambda_E(t) + \lambda_E(v). \tag{I.3.1}$$

Indeed, $E \setminus (E+t+v)$ is contained in $(E \setminus (E+t)) \cup ((E+t) \setminus (E+t+v))$.

Lemma I.3.1 (Two types of swaps)**.** *For every $w, w' \in W$ and $x, y \in A$,*

$$\lambda_S((w-w')(x-y)) \le b_h = 2h-1. \tag{I.3.2}$$

For every weight a_i and every $x, y \in A$, we also have

$$\lambda_S(a_i(x-y)) \le n+h-2. \tag{I.3.3}$$

Proof. Zero increments are immediate. For the first inequality, reserve distinct x, y for two positions of different weights. The remaining $n-2$ nonzero weights run over $A \setminus \{x, y\}$, still with excess h. There are at least four variables and fewer than $n \le p-2$, so neither the two-variable correction nor the excluded dimension $p-1$ applies. Minimality gives at least $(n-2)h+1$ sums. Swapping the two reserved points produces two translates of this set inside S, separated by $(w-w')(x-y)$. Their overlap gives the claimed intersection bound, and $s \le nh$ proves (I.3.2).

For the second inequality, reserve x, y and let the other $n-1$ weights run over $A \setminus \{x, y\}$, with excess $h-1$. Minimality over all excesses gives at least $(n-1)(h-1)+1$ sums. Put either x or y in position i, leaving the other unused. The same intersection argument proves the result. □

These estimates concern actual product increments. The bound b_h has not been asserted for every vector in their span.

Lemma I.3.2 (Occupied lines)**.** *Let E be finite with $|E| < p$. For $t \neq 0$, let $b_t(E)$ be the number of affine $\mathbb{F}_p t$-lines meeting E. Then*

$$b_t(E) \le \lambda_E(t).$$

For independent t_1, t_2, we have $|E| \le b_{t_1}(E) b_{t_2}(E)$. For three independent directions,

$$|E|^2 \le b_{t_1}(E) b_{t_2}(E) b_{t_3}(E). \tag{I.3.4}$$

The ambient space may have arbitrary dimension; E need not lie in one coset of the span of these directions.

Proof. Every occupied p-point line contains a nonempty proper subset of E, and therefore contributes at least one boundary point. The two-direction assertion follows from the injection $z \mapsto (z + \mathbb{F}_p t_1, z + \mathbb{F}_p t_2)$.

For three directions, work first in one affine coset of their three-dimensional span, with coordinates (x, y, z). Write $f(x,y), g(y,z), k(x,z)$ for the indicators of the three coordinate projections, of sizes b_3, b_1, b_2. Put $G_y = \sum_z g(y,z)$ and $K_x = \sum_z k(x,z)$. Two applications of Cauchy–Schwarz give

$$|E_c| \le \sum_{x,y} f(x,y) \sqrt{G_y K_x}, \qquad |E_c|^2 \le |\operatorname{supp} f| \Big(\sum_y G_y \Big) \Big(\sum_x K_x \Big).$$

Hence $|E_c|^2 \le b_{1,c}b_{2,c}b_{3,c}$. Summing over the occupied cosets gives

$$|E|^2 \le \left(\sum_c b_{1,c}\right)\left(\sum_c b_{2,c}b_{3,c}\right) \le \prod_{i=1}^{3}\left(\sum_c b_{i,c}\right).$$

Direction lines partition according to these cosets, giving (I.3.4). □

Proposition I.3.3 (Dimension bound)**.** *Let* $M = \operatorname{span}_{\mathbb{F}_p}\{a_i(x-y) : 1 \le i \le n,\ x, y \in A\}$. *Then*

$$1 \le \dim_{\mathbb{F}_p} U \le 2, \qquad 1 \le \dim_{\mathbb{F}_p} L \le 2.$$

If $\dim_{\mathbb{F}_p} U = 2$*, then* $L = M = cU$ *for any nonzero weight difference* c*, and*

$$n \le 4h - 3. \tag{I.3.5}$$

In this case a translate of S *is contained in* M*.*

Proof. By Lemma I.2.1, the weights are not all equal. Choose $c \in W - W$ nonzero; then $0 \ne cU \subseteq L \subseteq M$. If $\dim_{\mathbb{F}_p} L \ge 3$, select three independent genuine generators. Lemmas I.3.1 and I.3.2 imply $s^2 \le b_h^3$. However, $s \ge K_0 \ge h^2 + h + 1$, and

$$(h^2+h+1)^2 - (2h-1)^3 = h^2(h-3)^2 + 6h^2 - 4h + 2 > 0,$$

a contradiction. Thus $\dim_{\mathbb{F}_p} L \le 2$ and $\dim_{\mathbb{F}_p} U \le 2$.

If $\dim_{\mathbb{F}_p} U = 2$, then $L = cU$. The two-direction line bound gives $K_0 \le b_h^2$, which simplifies to (I.3.5). If $\dim_{\mathbb{F}_p} M \ge 3$, select two independent genuine generators of L and a genuine generator of M outside L. Then

$$s^2 \le (2h-1)^2(n+h-2).$$

Put $v = n - h - 2 \ge 0$. The difference from K_0^2 is

$$\begin{aligned} K_0^2 - (2h-1)^2(n+h-2) &= h^2(h-3)^2 + 2h^2 + 1 \\ &\quad + v(2h^3 - 2h^2 + 6h - 1) + v^2h^2 > 0, \end{aligned}$$

again a contradiction. Hence $M = L = cU$. Fixing $\alpha \in A$, each sum minus $\alpha\sum_i a_i$ equals $\sum_i a_i(x_i - \alpha) \in M$, proving the last assertion. □

I.4 Compression with modular wraparound and integer triangulation

All vector-space identifications in this section concern addition only. Field multiplication is not transported by an arbitrary linear map, and no translation-boundary estimate is transferred to a compressed set. The notation jE means an unrestricted sum of j copies of E, allowing repetitions; it is not scalar dilation.

I.4.1 Cauchy–Davenport and compression of different summands

For nonempty finite X, Y in any field of characteristic p,

$$|X + Y| \ge \min(p, |X| + |Y| - 1). \tag{I.4.1}$$

Choose subsets of sizes a, b with $a + b - 1 = \min(p, |X| + |Y| - 1)$. If their sumset has at most $a + b - 2 = d < p$ elements, choose a monic degree-d polynomial Q vanishing

there. The polynomial $Q(x+y)$ vanishes on the product grid but has nonzero coefficient $\binom{d}{a-1}$ at $x^{a-1}y^{b-1}$, contradicting the interpolation formula in Appendix A. Iteration gives $\min(p, \sum_i |X_i| - r + 1)$ for r summands.

In $\mathbb{F}_p^2$, horizontal compression C_x replaces every nonempty horizontal fiber by the initial interval of the same size beginning at zero. For arbitrary, possibly equal, finite sets $A_1, \ldots, A_r$,

$$|C_x A_1 + \cdots + C_x A_r| \le |A_1 + \cdots + A_r|. \tag{I.4.2}$$

Fix a target row and a tuple of source rows whose indices sum to it. The bound in (I.4.1) equals the size of the corresponding compressed fiber sum, an initial interval or all of $\mathbb{F}_p$. All such intervals in the target row are nested. Their union therefore has the largest of their sizes, while the original target row contains a fiber sum of at least that size. Sum over target rows. Vertical compression C_y is analogous. Modular wraparound is allowed in this argument.

I.4.2 Preserving dimension and compressing opposite sets

If A is noncollinear, choose a direction determined by two of its points as horizontal. Some row has at least two points, and at least two rows are occupied. Two compressions give

$$B = C_y C_x A = \{(x,t) : 0 \le t < r_x\}, \qquad r_x = \#\{y : |A_y| > x\}.$$

The numbers r_x are nonincreasing, so B is a Ferrers downset in the nonnegative integer quadrant. It contains $0, e_1, e_2$, has m points, and each coordinate is at most $m-1$. The multisets of row sizes of A and $-A$ agree, so

$$C_y C_x(-A) = B \tag{I.4.3}$$

as an exact equality of sets. Applying (I.4.2) to k copies each of A and $-A$ gives

$$|k(A-A)| \ge |2kB|.$$

I.4.3 An integer planar sumset bound

For any noncollinear integer point set B of size m,

$$|jB| \ge L_j(m) := \frac{j(j+1)}{2}m - (j^2-1) \qquad (j \ge 1). \tag{I.4.4}$$

Triangulate its convex hull using all points of B: triangulate the boundary polygon and insert each remaining point into its containing face or edge. If b, e, t count boundary vertices, edges, and triangles, Euler's formula and edge–face counting give

$$e = 3m - b - 3, \qquad t = 2m - b - 2.$$

The set jB contains the m dilated vertices, $j-1$ positive integer combinations in each dilated edge interior, and $\binom{j-1}{2}$ in each dilated triangle interior. Relative interiors are disjoint, and affine independence makes the combinations in each face distinct. Thus

$$|jB| \ge m + (j-1)e + \binom{j-1}{2}t = j^2 m - \frac{j(j-1)}{2}b - j^2 + 1 \ge L_j(m).$$

For $j = 1, 2$ the triangle-interior contribution is zero.

If $j(m-1) < p$, every coordinate of the integer set jB is smaller than p, so reduction modulo p is injective. Compression and (I.4.4) therefore yield

$$|jA| \ge L_j(m) \qquad \text{if } j(m-1) < p, \tag{I.4.5}$$

$$|k(A-A)| \ge L_{2k}(m) \qquad \text{if } 2k(m-1) < p. \tag{I.4.6}$$

The absence of wraparound is used only for the compressed integer set. No rectification assumption is made on the original A.

I.5 Correlation counts and exclusion of unbounded parameters

Suppose now that a minimal counterexample has $\dim_{\mathbb{F}_p} U = 2$. By Proposition I.3.3, after choosing a nonzero weight difference c, we work in the actual additive plane $M = cU$, containing a translate of S. For a finite increment set $X \subseteq M$, we use

$$\sum_{t \in M} |S \cap (S + t)| = s^2. \tag{I.5.1}$$

This counts ordered pairs of points of S. If $\lambda_S(t) \le d < s$ for every $t \in X$, then

$$|X|(s - d) \le s^2. \tag{I.5.2}$$

Fix $a_0 \in A$. Every increment in $j(cA - ca_0)$ is a sum of j genuinely controlled differences. Subadditivity gives boundary at most jq. By (I.4.5), if $j(m-1) < p$ and $jq < s$,

$$L_j(m)(s - jq) \le s^2. \tag{I.5.3}$$

Likewise, if $2k(m-1) < p$ and $kq < s$, (I.4.6) gives

$$L_{2k}(m)(s - kq) \le s^2. \tag{I.5.4}$$

Proposition I.5.1 (Finite parameter bound)**.** *A minimal counterexample with* $\dim_{\mathbb{F}_p} U = 2$ *must satisfy*

$$h \le 21, \qquad n \le 23, \qquad m \le 44, \qquad p < 736.$$

Proof. Define the concave quadratic

$$G_j(z) = L_j(m)(z - jq) - z^2.$$

If $j(m-1) \le K_0$ and $jq < K_0$, the hypotheses of (I.5.3) hold throughout $K_0 \le s \le nh$. If $G_j(K_0) > 0$ and $G_j(nh) > 0$, concavity makes G_j positive on this whole interval, contradicting (I.5.3).

First exclude all $h \ge 22$. Write uniquely

$$h = 3u + H, \quad H \in \{22, 23, 24\},\ u \ge 0, \quad n = h + 2 + v,\ v \ge 0, \quad j = u + 8.$$

For each H, the following four polynomials have nonnegative coefficients and strictly positive constant terms:

$$K_0 - j(m-1), \qquad K_0 - jq, \qquad 2G_j(K_0), \qquad 2G_j(nh) \tag{I.5.5}$$

The exact expansions are reconstructed by the ancillary verifier described in Section IV.2. The first two polynomials for the three values of H are

$$\begin{aligned}
H = 22: &\quad 3u^2 + 2uv + 42u + 14v + 147, \quad 3u^2 + 3uv + 44u + 22v + 163;\\
H = 23: &\quad 3u^2 + 2uv + 46u + 15v + 177, \quad 3u^2 + 3uv + 48u + 23v + 193;\\
H = 24: &\quad 3u^2 + 2uv + 50u + 16v + 209, \quad 3u^2 + 3uv + 52u + 24v + 225.
\end{aligned}$$

The constant terms of the two endpoint polynomials are, respectively, $(5220, 28656)$, $(31072, 54700)$, and $(59248, 82800)$. These coefficient identities establish positivity for all $u, v \ge 0$. Consequently $h \le 21$.

For $h \le 6$, (I.3.5) already gives $n \le 21$. For $h = 7, \dots, 21$, put $n = 24 + v$ with $v \ge 0$ and choose, respectively,

$$j = 4, 4, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 7, 7, 7.$$

The same verifier expands these fifteen univariate instances of (I.5.5), again obtaining nonnegative coefficients and positive constant terms. This excludes every $n \geq 24$. There are $(3 + 15) \times 4 = 72$ identities in total. Hence $n \leq 23$ and $m \leq 44$. Finally, (I.2.3) gives

$$p < \frac{n(3n-5)}{2} \leq \frac{23(69-5)}{2} = 736.$$

□

I.6 Symmetric planar doubling and wraparound

Lemma I.6.1 (Symmetric doubling)**.** *Let $p \geq 5$ be prime and $T \subseteq \mathbb{F}_p^2$ satisfy $0 \in T = -T$ and span the plane. Then*

$$|T + T| \geq 3\min(|T|, p) - 3. \tag{I.6.1}$$

Proof. Take $t \neq 0$ in T as horizontal. The central row contains the distinct points $0, t, -t$. Choose $u \in T$ outside it; $u, -u$ lie in distinct nonzero rows. The twice-compressed Ferrers downset B thus has width and height at least three and contains

$$X = \{0, e_1, 2e_1, e_2, 2e_2\}.$$

Compression gives $|2T| \geq |2B|$. Put $M_0 = \min(|T|, p)$. If $|B| > p$, remove maximal elements outside X until p remain. This is possible because X is a downset and $p \geq |X| = 5$. The resulting downset B' contains X, has M_0 points, and satisfies $|2T| \geq |2B'|$.

Let its width and height, counted in points, be $a, b \geq 3$. Its first row and column give

$$a + b - 1 \leq M_0 \leq p. \tag{I.6.2}$$

If $2a - 2 < p$ and $2b - 2 < p$, there is no wraparound in the integer set $2B'$, and (I.4.4) gives $|2B'| \geq 3M_0 - 3$.

Otherwise, exchanging coordinates if necessary, assume $2a - 2 \geq p$. Since p is odd,

$$a \geq (p+3)/2, \qquad b \leq p - a + 1 \leq (p-1)/2.$$

Thus row indices do not wrap: $2b - 2 < p$. Both directions cannot wrap simultaneously. Write the row lengths as

$$a = \lambda_0 \geq \lambda_1 \geq \cdots \geq \lambda_{b-1} = \ell \geq 1, \qquad \sum_j \lambda_j = M_0.$$

The zeroth row added to itself covers all p horizontal residues. A sum involving exactly one positive row index has horizontal length at most $a + \lambda_1 - 1 \leq M_0 - 1 < p$; one involving two positive row indices has length at most $2(M_0 - a) - 1 \leq 2(p - a) - 1 < p$. These sums are initial integer intervals, so the following contributions are valid:

1. Row zero contributes p points.
2. For $j = 1, \ldots, b-1$, rows zero and j contribute at least $a + \lambda_j - 1$ points in row j.
3. For $j = 1, \ldots, b-1$, rows $b-1$ and j contribute at least $\ell + \lambda_j - 1$ points in row $b - 1 + j$.

These three ranges of target rows are disjoint and do not wrap. Hence

$$\begin{aligned} |2B'| &\geq p + 2(M_0 - a) + (b-1)(a + \ell - 2) \\ &\geq p + 2M_0 - 2 \geq 3M_0 - 2. \end{aligned}$$

The second inequality uses $b - 1 \geq 2$ and $\ell \geq 1$. This is stronger than required and completes the wraparound case. □

Return to the two-dimensional minimal counterexample. The core conditions give

$$K_0 - 2(m-1) = n(h-2) - 3h + 3 \geq 0. \tag{I.6.3}$$

For $h = 3$, the right side is $n - 6 \geq 0$; for $h \geq 4$, the bound $n \geq h + 2$ makes it at least $h^2 - 3h - 1 \geq 3$. Since $K_0 < p$, we get $2(m-1) < p$. Taking $k = 1$ in (I.4.6) gives

$$|A - A| \geq 3m - 3.$$

The genuine increment set $T = c(A - A)$ is symmetric, contains zero, spans the plane, and has boundary at most b_h at every element. Also,

$$K_0 - 2b_h = (n-5)h + 3 > 0.$$

Lemma I.6.1, subadditivity, and correlation counting give the necessary inequality

$$L_{\mathrm{sym}}(m,p)(s - 2b_h) \leq s^2, \qquad L_{\mathrm{sym}}(m,p) = 3\min(3m-3, p) - 3. \tag{I.6.4}$$

The truncation at p is essential. For $m = 10, p = 23$, for example, the coefficient is 66, rather than 78.

I.7 Endpoint certificates for the complete finite domain

By Proposition I.5.1, Lemma I.2.1, and (I.3.5), the parameters of a two-dimensional minimal counterexample lie in the following exact domain:

$$\begin{gathered} p \text{ prime}, \quad 2 \leq p < 736, \quad 6 \leq n \leq 23, \quad 3 \leq h \leq 21, \\ h + 2 \leq n \leq 4h - 3, \quad n < h(2h-1), \quad K_0 = (n-1)h + 1 < p, \\ p - n - h \geq \max\{8, (n-2)(h-1)\}, \\ 2p < n(3n-5), \quad p \leq nh + \binom{n}{2}, \\ h = n - 2 \implies p < (n-1)^2. \end{gathered} \tag{I.7.1}$$

These are proved necessary conditions; no stronger direction bound, multiplicity filter, or extra small-characteristic filter is used. In particular $p = 17$ is retained. Enumeration gives exactly 3,766 triples (p, n, h), whose intervals $K_0 \leq s \leq V = \min(nh, p-1)$ contain 39,541 integer values in total.

The ancillary verifier constructs a certificate for each triple of one of the following forms:

1. A symmetric certificate: $L = 3\min(3m-3, p) - 3$ and $d = 2b_h < K_0$, using (I.6.4).
2. A mixed certificate: a positive integer k with $2k(m-1) < p$ and $d = kq < K_0$, and $L = L_{2k}(m)$, using (I.5.4).

Each certificate consists of exact integers

$$g_0 = L(K_0 - d) - K_0^2 > 0, \qquad g_1 = L(V - d) - V^2 > 0. \tag{I.7.2}$$

Since $G(z) = L(z-d) - z^2$ is concave, strict positivity at both endpoints gives strict positivity throughout $[K_0, V]$, contradicting the relevant necessary inequality.

The computation finds 1,947 symmetric certificates, 565 mixed certificates with $k = 2$, and 1,254 with $k = 3$. The minimum endpoint margin is 21. For a directly checkable example involving truncation, take $(p, n, h) = (23, 6, 4)$:

$$m = 10, \quad K_0 = 21, \quad V = 22, \quad b_h = 7, \quad L = 66,$$

The endpoint margins are $66 \cdot 7 - 21^2 = 21$ and $66 \cdot 8 - 22^2 = 44$.

Every admissible parameter triple contradicts a proved inequality, so no two-dimensional minimal counterexample exists. Proposition I.3.3 gives

$$\dim_{\mathbb{F}_p} U = 1. \tag{I.7.3}$$

This finite verification is sufficient because the uniform polynomial identities of Section I.5 first exclude the entire unbounded region. The computation neither extrapolates from finitely many primes nor enumerates the original point sets or weights.

I.8 From a point line to an exact prime-field model

Write $A = \alpha + uB$ with $u \neq 0$ and $B \subseteq \mathbb{F}_p$, and put $D = B - B$. By (I.6.3) and $K_0 < p$, we have $p \geq K_0 + 1 \geq 2m - 1$. Thus (I.4.1) gives

$$|D| \geq 2m - 1 > 2n. \tag{I.8.1}$$

The endpoint $p = 2m - 1$ is included.

Proposition I.8.1 (A rectangle of genuine increments)**.** *Under* (I.7.3)*, we have* $\dim_{\mathbb{F}_p} L = 1$.

Proof. Since D contains a nonzero prime-field element, $L = \operatorname{span}_{\mathbb{F}_p}\{uc : c \in W - W\}$. If its dimension is at least two, choose actual weight differences c_1, c_2 such that $v_i = uc_i$ are independent. Put

$$X = v_1 D + v_2 D.$$

Independence makes $(d_1, d_2) \mapsto v_1 d_1 + v_2 d_2$ injective, so $|X| = |D|^2 > 4n^2$. Each $v_i d$ is a genuine increment $c_i(x - y)$, so every element of X has boundary at most $2b_h$. Even if S is not contained in this plane, summing intersections over X counts only some ordered pairs of S. Therefore

$$|X|(s - 2b_h) \leq s^2, \qquad |X| \leq \frac{(nh)^2}{K_0 - 2b_h}.$$

Here the numerator and denominator are bounded separately; no monotonicity assertion is needed. The exact identity

$$4(K_0 - 2b_h) - h^2 = 3(h-2)^2 + 4h(n-h-2) > 0$$

gives $|X| < 4n^2$, a contradiction. A nonzero weight difference and point difference ensure $L \neq 0$, proving the proposition. □

Choose a nonzero weight difference c and a base weight a. Since $\dim_{\mathbb{F}_p} L = 1$, write $a_i = a + ct_i$ with $t_i \in \mathbb{F}_p$. Translating the sumset and dividing by cu gives the equicardinal set

$$T = \left\{ \beta \sum_i x_i + \sum_i t_i x_i : x_i \in B \text{ pairwise distinct} \right\}, \qquad \beta = a/c.$$

If $\beta \notin \mathbb{F}_p$, the linear functional on $\operatorname{span}_{\mathbb{F}_p}\{1, \beta\}$ taking $1 \mapsto 0$ and $\beta \mapsto 1$ maps T onto the ordinary restricted n-fold sumset of B. Appendix C proves directly, for the present range $n \leq m < p$, that this image has at least $\min(p, nh + 1)$ elements. This contradicts $|T| = s < \min(p, nh + 1)$. Hence $\beta \in \mathbb{F}_p$, and all $a_i/c = \beta + t_i$ belong to $\mathbb{F}_p^*$.

This proves (I.1.2). The affine normalization of the point set and nonzero scaling of weights preserve all cardinalities, nonzero weights, equality classes, and multiplicities. Since n, p are unchanged, the exclusion $p \neq n + 1$ persists. Here $n \geq 6$, so both correction terms are zero.

If $\mathsf{EH}(\mathbb{F}_p)$ holds but some characteristic-p field has a counterexample, its minimal counterexample produces a prime-field counterexample by this exact model, a contradiction. The converse is immediate. Characteristic zero is covered by the Sun–Zhao theorem cited in Appendix A. This proves Theorem I.1.2.

Part II
Full insertions and capacity arguments over prime fields

II.1 Full insertions and actual sumsets

Throughout this part, F has prime characteristic p. Take $M < p$ weight slots, grouped by equal weights, with positive multiplicities $r_1, \dots, r_c$ and distinct values $u_1, \dots, u_c$. One value may be zero. Write

$$d = \sum_{i<j} r_i r_j, \quad W = \sum_i \binom{r_i}{2}, \quad B = \binom{M}{2} = d + W, \quad J_M = \prod_{j=0}^{M-1} j!.$$

The target degree is $0 \le k < p$ and the excess degree is $q = d - k$. The boundary parameter in Part I will henceforth be denoted $b_h = 2h - 1$ to avoid conflict with this q.

Let $b \in F^M$ be the slotwise weight vector, let $\Delta_M(v) = \prod_{a<b}(v_a - v_b)$, and let Δ_{in} be the product of the within-block Vandermonde polynomials. For formally distinct roots $x_1, \dots, x_M$, put

$$I(f) = \sum_{\sigma \in \mathfrak{S}_M} \frac{f(x_\sigma)}{\Delta_M(x_\sigma)}.$$

If the total degree is below B, then $I(f) = 0$; at degree B it is constant. Indeed, the numerator after taking a common denominator is alternating, hence divisible by Δ_M. For a degree-B monomial, a nonzero constant is possible only when the exponents permute $M-1, \dots, 0$, giving the corresponding sign.

Let T denote summation over ordered block partitions, with cross-block denominator $E = \prod_{i<j} \prod_{a \in i, b \in j}(x_a - x_b)$. If G is blockwise symmetric and homogeneous of degree $q \ge 0$, and $L = \sum_a b_a x_a$, monomial expansion and grouping within blocks give

$$T(GL^k) = \frac{k!}{J_M \prod_i r_i!} \big[\Delta_{\mathrm{in}}(\partial) G(\partial) \Delta_M\big](b). \tag{II.1.1}$$

Indeed, for a degree-$(B-k)$ monomial $P = x^\alpha$, the surviving exponents $\alpha + \beta$ in $I(PL_v^k)$ permute the staircase. Their coefficients are $k! v^\beta / \beta!$, whereas the corresponding coefficients in $P(\partial_v)\Delta_M(v)$ are $J_M v^\beta / \beta!$. Thus $I(PL_v^k) = (k!/J_M) P(\partial_v)\Delta_M$. Taking $P = \Delta_{\mathrm{in}} G$ proves the identity. All factorial arguments are smaller than p; neither $B!$ nor $d!$ is canceled.

Lemma II.1.1 (Full Taylor-layer criterion)**.** *Let $0 \le k < p$. Some G makes* (II.1.1) *nonzero if and only if the degree-$(B-k)$ homogeneous layer of $\Delta_M(b+z)$ is nonzero, equivalently $D_b^k \Delta_M \ne 0$, where $D_b = \sum_a b_a \partial_{z_a}$. Such a certificate guarantees at least $k+1$ permutation sums on every actual M-point set.*

Proof. Write $C_*(z)$ for this layer. Homogeneity and the Taylor formula with $k < p$ give $D_b^k \Delta_M(z) = k! C_*(z)$. The polynomial C_* is blockwise alternating, hence divisible by Δ_{in},

and has degree at most $M-1<p$ in each variable. The blockwise determinants

$$A_e(z)=\prod_i \det(z_{i,a}^{e_{i,b}})_{a,b}, \qquad M-1\ge e_{i,1}>\cdots>e_{i,r_i}\ge 0$$

form a basis of alternating polynomials. Differential coefficient pairing is diagonal on this basis, with nonzero diagonal entries $\prod_i r_i!\prod_b e_{i,b}!$, all units. If $C_*\neq 0$, select a basis element A_e pairing nontrivially with it and set $G=A_e/\Delta_{\mathrm{in}}$; then (II.1.1) is nonzero. The converse follows directly from that identity. If $k>d$, the layer has degree below W and is zero; no negative-degree insertion is introduced.

If an actual point set produced at most k permutation sums, choose a monic degree-k polynomial Q vanishing on them. Replacing L^k by $Q(L)$ leaves the nonzero constant unchanged, since T kills the lower-degree terms. Yet every specialized numerator is zero and every denominator is nonzero, a contradiction. □

In particular, for $k=d<p$, the ordinary volume is

$$T(L^d)=\frac{d!\prod_i\prod_{j=0}^{r_i-1} j!}{J_M}\prod_{i<j}(u_i-u_j)^{r_ir_j}\neq 0. \tag{II.1.2}$$

For any retained submultiset of slots, fix the points assigned to omitted slots and permute the retained slots on the remaining distinct points. A translate of the resulting sumset lies in the original sumset. Consequently every such lower bound lifts to the original problem. Each certificate may use its own fixed assignment; there is no prescribed finite list of insertions.

For subsequent calculations, factor

$$\Delta_M(b+z)=\Delta_{\mathrm{in}}(z)V(u)C(z),\quad V(u)=\prod_{i<j}(u_i-u_j)^{r_ir_j},\quad C(z)=\prod_{\substack{a<b\\ b_a\neq b_b}}\left(1+\frac{z_a-z_b}{b_a-b_b}\right).$$

Thus the required criterion is $C_q\neq 0$. For $q=1$, failure of all certificates is equivalent to

$$A_i:=\sum_{j\neq i}\frac{r_j}{u_i-u_j}=0 \quad \text{for every } i. \tag{II.1.3}$$

Equivalently, $\partial_i V=r_iVA_i=0$. Euler's identity always gives

$$\sum_i r_iu_iA_i=d. \tag{II.1.4}$$

II.2 Matching formulas and small excess degrees

Assume $M<p$, $0\le k<p$, and $q=d-k\ge 0$; at each application we also verify $q<p$. The matching lemmas require $q\ge 2$, whereas the singleton lemma also allows $q=0,1$. In the squarefree quotient $z_a^2=0$, the formal logarithm through degrees below p gives

$$\log C=\sum_a A_{i(a)}z_a+\sum_{\substack{a<b\\ i(a)\neq i(b)}}\frac{z_az_b}{(u_{i(a)}-u_{i(b)})^2}.$$

Every squarefree coefficient is therefore a matching sum on the chosen slots. An unmatched vertex in block i contributes A_i; a cross-block edge contributes $c_{ij}=(u_i-u_j)^{-2}$; a within-block edge has weight zero. Direct expansion of the factors gives the same identity, so no analytic logarithm is involved.

II.2.1 Two large blocks and asymmetric odd degrees

Select a, b slots from two blocks, and write $A = A_1$, $B = A_2$, $c = (u_1 - u_2)^{-2}$. The matching coefficient is

$$H_{a,b} = \sum_{s=0}^{\min(a,b)} \binom{a}{s}\binom{b}{s} s! c^s A^{a-s} B^{b-s}. \tag{II.2.1}$$

All factorial arguments are below p. Although their integer products may exceed p, these combinatorial factors remain units. Define

$$L_t^{\alpha}(x) = \sum_{j=0}^{t} \binom{t+\alpha}{t-j} \frac{(-x)^j}{j!}.$$

For $a \geq b$, coefficient comparison gives $H_{a,b} = b! c^b A^{a-b} L_b^{a-b}(x)$, where $x = -AB/c$. This remains a polynomial identity when $A = 0$. We need the following three elementary identities:

$$(L_t^{\alpha})' = -L_{t-1}^{\alpha+1}, \tag{II.2.2}$$

$$x(L_t^{\alpha})'' + (\alpha + 1 - x)(L_t^{\alpha})' + tL_t^{\alpha} = 0, \tag{II.2.3}$$

$$L_{t-1}^{\alpha+2} = (L_t^{\alpha})'' - (L_t^{\alpha})'. \tag{II.2.4}$$

They follow coefficientwise from binomial identities, using $\alpha = 0, 1$. A polynomial of degree $t < p$ satisfying this differential equation cannot have a multiple nonzero root: at a root x_0 of multiplicity $2 \leq \ell \leq t$, the lowest nonzero coefficient would be $x_0 \ell(\ell - 1) \neq 0$. Also $L_t^0(0) = 1$ and $L_t^1(0) = t + 1 \neq 0$. Hence at a nonzero root,

$$L_{t-1}^{\alpha+2}(x) = -(\alpha + 1)(L_t^{\alpha})'(x)/x \neq 0.$$

When $t = 1$, the second derivative is zero and the assertion is checked directly.

Lemma II.2.1 (Two-block certificates)**.** *Let $q \geq 2$. Any of the following conditions ensures $C_q \neq 0$:*

1. *$q = 2t$, with two blocks of sizes at least $t + 1, t$.*
2. *$q = 2t + 1$, $t \geq 1$, with two blocks of sizes at least $t + 2, t + 1$, and $d \neq 0$ in F.*
3. *$q = 2t + 1$, $t \geq 1$, with two blocks of sizes at least $t + 2, t$, at least $t + 2$ slots outside the first block, and $d \neq 0$ in F.*

Proof. Suppose $C_q = 0$. In the even case, selecting (t, t) gives $L_t^0(x) = 0$, hence $A, B, x \neq 0$. Selecting $(t + 1, t - 1)$ also gives $L_{t-1}^2(x) = 0$, contradicting the preceding identities.

In the third case, selection $(t + 1, t)$ gives $AL_t^1(x) = 0$. If $A \neq 0$, then $x, B \neq 0$, and selection $(t + 2, t - 1)$ forces $L_{t-1}^3(x) = 0$, a contradiction. Thus $A = 0$. Select t slots from the first block and any $t + 1$ outside slots, forming S. Every surviving matching must match all first-block vertices, so its coefficient is

$$t! \prod_{a \in S} (u_1 - b_a)^{-2} \sum_{a \in S} A_{i(a)} (u_1 - b_a)^2.$$

There are at least $t + 2$ outside slots. Comparing subsets differing in one element makes all summands equal; since $t + 1$ is a unit, all are zero. Hence every $A_i = 0$, contradicting $d \neq 0$ in (II.1.4).

The same argument proves the second case except possibly when there are only two blocks and exactly $t + 1$ outside slots. In general, selections $(t + 1, t)$, $(t, t + 1)$, and $(t + 2, t - 1)$, together with the same multiple-root argument, give $A = B = 0$. For any third block j, selection $(t, t, 1)$ has surviving coefficient $A_j t! c^t$, so $A_j = 0$. With only two blocks all A_i already vanish. Euler's identity again contradicts $d \neq 0$. □

Lemma II.2.2 (One large block)**.** *Let $q \geq 2$. Suppose a block has size $H \geq q$ and there are N slots outside it. If $q = 2t$ and $N \geq t$, or if $q = 2t+1$, $N \geq t+2$, and $d \neq 0$, then $C_q \neq 0$.*

Proof. If $C_q = 0$, selection of q slots inside the block gives $A^q = 0$, hence $A = 0$. For even q, select t inside and t outside slots; the coefficient is $t!$ times a product of nonzero edge weights, hence nonzero. For odd q, use the $(t, t+1)$ subset comparisons from the preceding proof to obtain all $A_i = 0$, contradicting Euler's identity. □

II.2.2 An odd balanced certificate with three blocks

Lemma II.2.3 (Odd balanced case)**.** *If $q = 2t+1$, $t \geq 1$, there are at least three distinct blocks, two have size at least $t+1$, and $d \neq 0$, then $C_q \neq 0$.*

Proof. Write $a = A_1$, $b = A_2$, $c = (u_1 - u_2)^{-2}$, $y = ab$, and define

$$F(y) = \sum_{j=0}^{t} \binom{t}{j}^2 j! c^j y^{t-j}, \quad Q(y) = \sum_{j=0}^{t} \binom{t+1}{j}\binom{t}{j} j! c^j y^{t-j}.$$

Coefficient comparison gives

$$cyF'' + (c+y)F' - tF = 0, \quad cyQ'' + (2c+y)Q' - tQ = 0, \quad Q = F + cF'. \tag{II.2.5}$$

Their constant terms are $t!c^t$ and $(t+1)!c^t$. Their nonzero roots are simple by the same differential-equation argument.

If $C_q = 0$, selections $(t+1, t)$ and $(t, t+1)$ give $aQ(ab) = bQ(ab) = 0$. Since $Q(0) \neq 0$, either $a = b = 0$ or both are nonzero. In the first case, selection $(t, t, 1)$ forces $A_j = 0$ for each third block, contradicting Euler's identity.

Thus $a, b, y \neq 0$, $Q(y) = 0$, and $Q'(y) \neq 0$. Moreover $F' = -F/c$ and $F \neq 0$, since otherwise F, F' vanish together, contrary to simplicity from the first differential equation. Fix a third block with $A_j = A$ and edge weights $e = (u_j - u_1)^{-2}$ and $g = (u_j - u_2)^{-2}$. Selection $(t, t, 1)$ has coefficient $AF + (be + ag)F'$, so

$$A = (be + ag)/c. \tag{II.2.6}$$

For $t \geq 2$, select $(t+1, t-1, 1)$. Termwise differentiation of $H_{t+1,t} = aQ(ab)$ and $H_{t,t} = F(ab)$ gives precisely

$$D = \frac{a^2 Q'}{t} A + \frac{(t+1)aF'}{t} e + \frac{a^3 Q''}{t} g.$$

At $Q(y) = 0$, (II.2.5) gives $Q''/Q' = -(2c+y)/(cy)$ and $yQ' = -(t+1)cF'$. Dividing $D = 0$ by the nonzero factor $a^2 Q'/t$ gives $A = be/c + (2c+y)g/(bc)$. Subtracting (II.2.6) gives $2g/b = 0$, a contradiction.

For $t = 1$, $F = y + c$, $Q = y + 2c$, and $y = -2c$. Instead select $(2, 0, 1)$, with coefficient $D = Aa^2 + 2ea$. Substitution from (II.2.6) gives $D = a^3 g/c \neq 0$, avoiding negative subscripts. All divisors are units because $M \geq 2t+3$ and $p > M$. □

II.2.3 A characteristic operator for singleton blocks

Lemma II.2.4 (Singleton certificate)**.** *Let $S_0 = \sum_{r_i \geq 2} r_i$ count slots in nonsingleton blocks. If $S_0 \leq q \leq M-1$, then $C_q \neq 0$.*

Proof. Every monomial of the Vandermonde polynomial has exponent set $0, \ldots, M-1$. Hence

$$\sum_a v_a^q \partial_a^q \Delta_M = \lambda_q \Delta_M, \qquad \lambda_q = q! \binom{M}{q+1} \neq 0.$$

Replacing v_a^q by any monic degree-q polynomial $f(v_a)$ preserves the identity: lower terms give alternating polynomials of degree below B, hence zero. Take $f(T) = T^{q-S_0}\prod_{r_i\geq 2}(T-u_i)^{r_i}$, apply $P = \Delta_{\text{in}}(\partial)$, and evaluate at $v = b$. The differentiation order in a slot is at most $r_i - 1$, so the multiplicity-r_i zero of f kills every nonsingleton-block term. The singleton terms give

$$\sum_{a \text{ a singleton slot}} f(b_a)[P\partial_a^q\Delta_M](b) = \lambda_q[P\Delta_M](b) \neq 0.$$

The final factor on the right is $\prod_i(r_i!\prod_{j<r_i} j!)V(u)$. Thus some allowable degree-q insertion is nonzero, and Lemma II.1.1 applies. □

II.2.4 Even boundary cases and terminal degrees three and four

Lemma II.2.5 (Even boundary)**.** *If $q = 2t$, $t \geq 2$, two blocks have size at least t, and there are at least five distinct blocks, then $C_q \neq 0$.*

Proof. If every coefficient vanished, selection (t, t) would force $F = L_t^0(x) = 0$, with $x = -AB/c$, so $A, B, x, F' \neq 0$. Fix a third block j and write $a = (u_1 - u_j)^{-2}$ and $b = (u_2 - u_j)^{-2}$. Selection $(t, t-1, 1)$ has coefficient

$$A_jH_{t,t-1} + taH_{t-1,t-1} + (t-1)bH_{t,t-2}.$$

By (II.2.1), the three H terms equal $-(t-1)!c^{t-1}AF'$, $(t-1)!c^{t-1}L_{t-1}^0$, $(t-2)!c^{t-2}A^2F''$. Using $xF' = t(F - L_{t-1}^0)$ and $xF'' + (1-x)F' + tF = 0$, divide by the known nonzero common factor to obtain

$$-AA_j - xa + (A^2/c)b(x-1)/x = 0.$$

Exchanging the first two blocks and eliminating A_j gives $Ab - Ba = 0$. Every third-block value must therefore be a root of $A(u_1 - T)^2 - B(u_2 - T)^2$. This is a nonzero polynomial of degree at most two: if its quadratic term vanishes, its linear term is $2A(u_2 - u_1) \neq 0$. At least five blocks supply three distinct roots, a contradiction. □

Call a positive multiplicity vector strictly terminal under slot deletion at target k if $0 \leq q = d - k < M - H$, where $H = \max r_i$. Its degree is at least k, and deleting any positive slot drops the degree below k.

Lemma II.2.6 (Saturated degrees three and four)**.** *Suppose $k = p - 1$, the configuration is strictly terminal under slot deletion, and $q = 3$ or 4. Then $C_q \neq 0$.*

Proof. For $q = 3$, if there are two nonsingleton blocks and at least three blocks, apply Lemma II.2.3 with $t = 1$. With only two blocks, terminality forces both sizes to be at least four, so Lemma II.2.1 applies. If just one block is nonsingleton, use Lemma II.2.2 when $H \geq 3$, and Lemma II.2.4 when $H = 2$. All-singleton configurations use the latter too. Here $d = p + 2 \neq 0$.

For $q = 4$, use one large block if $H \geq 4$. If $H = 3$ and another nonsingleton block exists, use two blocks $(3, 2)$; otherwise $S_0 \leq 4$, giving the singleton certificate. If $H = 2$ and at most two blocks are nonsingletons, again $S_0 \leq 4$. With at least three blocks of size two and at least five blocks in total, use the even boundary lemma. With at most four blocks, terminality forces $M > 6$, leaving only $(2, 2, 2, 1)$ and $(2, 2, 2, 2)$. Their degrees are $18, 24$; since $d = p + 3$, they would require the nonprimes $p = 15, 21$. If $H = 1$, use the singleton certificate. All singleton applications satisfy $q < M - H \leq M - 1$. □

Lemma II.2.7 (A nonsaturated rescue at degree equal to the characteristic)**.** *Any positive multiplicity configuration with $M < p$ and $d = p$ produces at least $p - 1$ permutation sums on every actual M-point set.*

Proof. Use the stronger target $k' = p-2$, so the excess degree is two. If a nonsingleton block exists, $d > 0$ ensures another block, and Lemma II.2.1 applies to $(2,1)$. For all singleton blocks, $M \geq 3$ and Lemma II.2.4 applies. Both give $k'+1 = p-1$ sums. Thus any original target $K \leq p-2$ is met. This is a stronger cardinality conclusion after changing target, rather than an assertion about the same algebraic certificate. □

Lemma II.2.8 (Source constraints with at most one nonsingleton block)**.** *Suppose the original core satisfies $h \geq 3$, $n \geq h+2$, $(n-1)h < p-1$, and all capacities are at most h. A saturated configuration retained from it, strictly terminal under slot deletion and with at most one nonsingleton block, has a certificate for the original target.*

Proof. Use volume for $q = 0$ and Lemma II.2.4 for all singleton blocks. If the only nonsingleton block has size H, use that lemma for $q \geq H$, and one large block with terminality for $2 \leq q \leq H$. It remains to exclude $q = 1, d = p$. If there are s singleton blocks, $p = Hs + \binom{s}{2} = s(2H+s-1)/2$. Terminality gives $s > 1$, while $s < M < p$. Primality forces $s = 2$, hence $p = 2H+1 \leq 2h+1$. But the source conditions give $p > (n-1)h+1 \geq h(h+1)+1 > 2h+1$, a contradiction. □

II.3 Swap batches and a linear parameter bound

The additional published input here is Balandraud's theorem on subset sums of an opposite-free set [3, Theorem 5]: if p is odd and $D \subseteq \mathbb{F}_p$ satisfies $D \cap (-D) = \varnothing$, then its subset sums, including the empty sum, satisfy

$$|\Sigma(D)| \geq \min\left(p, 1 + \frac{|D|(|D|+1)}{2}\right). \tag{II.3.1}$$

No zero-sum-free assumption is added. The assertion includes saturation to all of $\mathbb{F}_p$.

Lemma II.3.1 (Swap-batch lower bound)**.** *Let $A \subseteq \mathbb{F}_p$ have $m \leq p$ points, and let each value among m weights have multiplicity at most $\lceil m/2 \rceil$. Put $\epsilon = m \bmod 2$. Their permutation sumset has at least*

$$\min\left(p, 1 + \left\lceil \frac{m^2 + 2m - 3\epsilon}{16} \right\rceil\right)$$

elements.

Proof. First pair $2\lfloor m/2 \rfloor$ weight slots with unequal values. For $m = 2r$, order equal values consecutively and pair positions $j, j+r$. An equal-valued pair would require a consecutive block of length at least $r+1$, impossible. For odd m, first remove one slot from a largest class, so the remaining multiplicities are at most $(m-1)/2$, and pair as above, reserving the removed slot. Write $c_j \neq 0$ for the differences of the prescribed pairs.

With $R \geq 2$ points left, take $t = \lfloor (R+1)/3 \rfloor$ weight pairs. Choose disjoint point pairs successively so that $c_j(x_j - y_j)$ are nonzero and pairwise distinct up to sign within this batch. After $j < t$ choices, $R-2j$ points remain. Cauchy–Davenport gives at least $\min(p, 2R-4j-1)$ differences. Since $3j \leq R-2$, this exceeds the $2j+1$ forbidden increments, so the next pair exists.

Repeat batches until ϵ points remain and assign them to the reserved slot if needed. Starting from a fixed permutation, independent swaps of disjoint pairs produce a translate of a subset-sum set inside the actual support. Each batch satisfies (II.3.1). Increments in different batches may repeat or be opposites; applying Cauchy–Davenport to the actual batch subset-sum sets yields $\min(p, 1 + \sum t(t+1)/2)$.

For a batch taking R to $R' = R - 2t$, the exact identity is

$$\frac{t(t+1)}{2} - \frac{R^2 + 2R - R'^2 - 2R'}{16} = \frac{t(3t+1-R)}{4} \geq 0.$$

because $R \in \{3t-1, 3t, 3t+1\}$. Sum down to $R = \epsilon$ and round up. □

Apply the lemma to the exact prime-field model. Each zero-padded multiplicity is at most h, and $n \geq h+2$ gives $h < m/2$. A counterexample must therefore satisfy

$$(n+h)^2 + 2(n+h) - 3\epsilon < 16nh, \qquad \boxed{n < 14h}.$$

For the last implication, put $n = 14h + u$, $u \geq 0$; then $(n+h)^2 - 16nh = h^2 + 14hu + u^2 > 0$. No new minimality assumption is made over the prime field; only the preserved multiplicities and numerical constraints are used.

II.4 Globally minimal degree and the capacity frontier

Fix an original core. Sort its zero-padded capacities as $c_1 = h \geq c_2 \geq \cdots \geq c_t \geq 1$, of total $m = n + h$; break ties by placing the actual zero-weight class first. Put

$$K = \min(nh, p-1), \quad d(r) = \sum_{i<j} r_i r_j, \quad 0 \leq r_i \leq c_i.$$

Among integer vectors with $d(r) \geq K$, choose one of minimum degree, denoted $d = K + q$, with total M and largest multiplicity H. Such a choice exists because the full vector has degree at least nh. Deleting a positive slot lowers degree by $M - r_i > 0$, so global minimality gives

$$0 \leq q < M - H < p, \qquad d < 2p. \tag{II.4.1}$$

Every capacity continues to represent its original coefficient value. Equal-valued classes are never split.

Lemma II.4.1 (Prefix and fixed-total frontier)**.** *If $q > H$, one can choose a feasible vector of the same degree of the form*

$$r = (h, c_2, \ldots, c_{j-1}, s, 0, \ldots, 0), \qquad 1 \leq s \leq c_j,$$

where $H = h$, $N = M - h > q > h$, $j \geq 3$, and every feasible vector of total $M-1$ has degree below K.

Proof. Sorting a feasible multiplicity vector decreasingly leaves it feasible for the decreasing capacities. Otherwise its ith entry would exceed c_i, requiring at least i capacities strictly larger than c_i, when there are at most $i-1$. Sorting preserves degree and total, and represents a new feasible choice on the original labels; actual coefficient values are not exchanged.

Let j be the last positive entry of a sorted vector. If some $i < j$ is only partially retained, transfer a slot from j to i. The degree decreases by $r_i - r_j + 1$, between 1 and $H < q$, and still reaches K, contrary to minimality. Thus the vector is a prefix. Terminality gives $N > q > h$. Two blocks would give $N \leq h$, so $j \geq 3$.

Deleting a slot gives a total-$(M-1)$ vector of degree below K. Any two feasible vectors of that total are connected by transfers from surplus to deficient coordinates. An upward degree change is at most $h-1$. If some such vector reached K, the first crossing along a connecting path would have degree at most $K + h - 2 < K + q$, again contradicting minimality. □

Lemma II.4.2 (Total size and dual cost)**.** *In the branch $q > H$, we have $M < 6h$. Let $a \in \{0, \dots, h-1\}$ be the water level at total $M-1$, defined by*

$$\sum_i \min(c_i, a) \le M - 1 < \sum_i \min(c_i, a+1).$$

Put $\delta = h - a$, $\phi_a(x) = \binom{x}{2} - ax$, and

$$\psi_i = \phi_a(r_i) - \min_{0 \le x \le c_i} \phi_a(x), \qquad C = \sum_i \psi_i.$$

Then each $\psi_i \ge 0$, and

$$C \le N + \delta - q - 2, \quad a + \binom{\delta}{2} \le N - 3, \quad \delta^2 - 3\delta \le 8h - 8. \tag{II.4.2}$$

Proof. Since $\sum r_i^2 \le HM$, we have $d \ge M(M-H)/2$. Combining $d = K + q < K + M - H$ gives $(M-H)(M-2) < 2K \le 2nh$. If $h \ge 4$ and $M \ge 6h$, the left side is at least $5h(6h-2) \ge 28h^2 - 2h \ge 2nh$, using $n \le 14h - 1$, a contradiction. For $h = 3$, the earlier bound $n < 2h^2 - h = 15$ gives $M \le m < 18 = 6h$ directly.

At fixed total, maximizing the cross-block degree is equivalent to minimizing $W(r) = \sum_i \binom{r_i}{2}$: choose the cheapest $M-1$ entries from the capacity increment lists $0, 1, \dots, c_i - 1$. An optimal vector y has entries $\min(c_i, a)$, with some larger-capacity entries increased by one. Each entry minimizes ϕ_a. Hence $W(r) - W(y) = a + C$. The frontier lemma gives

$$d(r) - d(y) = M - 1 - a - C = N + \delta - 1 - C \ge q + 1,$$

which proves the first inequality. The first-block cost is $\binom{\delta}{2}$; use $q \ge h + 1$ for the second inequality. Expanding it with $N \le 5h - 1$ gives the third. □

For use in both height ranges, the cost of a full block x is

$$F_a(x) = \binom{\max(x - a, 0)}{2}.$$

and the cost for capacity c with s retained slots is

$$P_a(c, s) = \binom{s}{2} - as - \binom{t}{2} + at, \qquad t = \min(c, a). \tag{II.4.3}$$

An entirely omitted capacity x costs $ax - \binom{x}{2}$ when $x \le a$, and $\binom{a+1}{2}$ otherwise.

II.5 Dual-cost bootstrap for heights at least twenty-five

Keep the prefix notation in the branch $q > H$, and put

$$z = c_j - s, \quad R_0 = \sum_{i>j} c_i, \quad R = z + R_0 = m - M, \quad \beta = \frac{a(a+1)}{2h}.$$

The degree outside the first block is at least $N(N-h)/2$. Since $K \le nh = h(N+R)$, it is at most $hR + q < hR + N$. Thus

$$N(N - h - 2) < 2hR, \quad N \le h + 1 + 2R, \quad C - \delta \le 2R - 2. \tag{II.5.1}$$

The second inequality follows from $N > h$, including $R = 0$; the third also uses (II.4.2).

The first-block cost is $\psi_1 = \binom{\delta}{2}$. Together the first and last retained blocks satisfy

$$\psi_1 + \psi_j - \delta \geq f(z), \qquad f(z) = \left\lfloor \frac{(z-1)^2}{4} \right\rfloor - 1. \tag{II.5.2}$$

If $c_j \leq a$, then $\psi_j = z(a - c_j) + \binom{z+1}{2} \geq \binom{z+1}{2}$, and $\binom{\delta}{2} - \delta \geq -1$ proves the claim. If $c_j \geq a+1$ and $s \geq a$, then $z \leq \delta$ and $\psi_j \geq 0$. If $s < a$, put $u = a - s$, so $z \leq \delta + u$ and $\psi_j = \binom{u+1}{2}$. Use $u = 0$ in the preceding case. For fixed $w = \delta + u \geq 1$,

$$\min_{1 \leq \delta \leq w} \left\{ \binom{\delta}{2} - \delta + \binom{w - \delta + 1}{2} \right\} = \left\lfloor \frac{(w-1)^2}{4} \right\rfloor - 1.$$

The right side is nondecreasing on nonnegative integers, proving (II.5.2). Each omitted capacity $x \leq h$ costs at least βx: for $x \leq a$, use $ax - \binom{x}{2} \geq x(a+1)/2 \geq \beta x$; for $x \geq a+1$, use $x \leq h$. Consequently

$$(\beta - 2)R_0 \leq g(z), \qquad g(z) = 2z - 1 - \left\lfloor \frac{(z-1)^2}{4} \right\rfloor. \tag{II.5.3}$$

Lemma II.5.1 (The height-twenty-five cutoff)**.** *If $h \geq 25$, the branch $q > H$ has a certificate for the original target unless $d = p$ and $K \leq p - 2$. In that exception Lemma II.2.7 gives a stronger cardinality bound.*

Proof. The quadratic in (II.4.2) gives $\delta < h/2 + 3$: substitution of $h/2 + 3$ into its left side, minus the right side, is $(h^2 - 26h + 32)/4 > 0$ for $h \geq 25$. Hence $a > h/2 - 3 > 0$. For $h \geq 26$, $(h/2 - 3)(h/2 - 2) - 4h = (h^2 - 26h + 24)/4 > 0$, so $a(a+1) > 4h$. For $h = 25$, integrality gives $a \geq 10$ and the same conclusion. Thus $\beta > 2$.

The integer function g is nonnegative exactly when $1 \leq z \leq 9$, and has maximum five. Equation (II.5.3) forces this range. Nonnegativity of the other costs also gives

$$\frac{\delta(\delta - 3)}{2} \leq C - \delta - \beta R_0 \leq 2z - 2 - (\beta - 2)R_0 \leq 16.$$

Thus $\delta \leq 7$ and $a \geq h - 7$, whence $\beta \geq (h-7)(h-6)/(2h) > 6$; the last inequality is equivalent to $h^2 - 25h + 42 > 0$. Hence $R_0 \leq 1$. If $R_0 = 1$, then $g(z) > 4$, forcing $z = 4, 5, 6$; if $R_0 = 0$, we still have $z \leq 9$. Either way $R \leq 9$.

Since $\psi_1 - \delta \geq -1$, we also have $\psi_j \leq 2z - 1 - (\beta - 2)R_0 \leq 2z - 1$. If $c_j \leq a$, the explicit cost formula gives $z(a - c_j) + \binom{z+1}{2} \leq 2z - 1$. This excludes $z \geq 3$, while $z = 1, 2$ forces $c_j \geq a$. Thus always $c_j \geq a \geq h - 7$. Since $j \geq 3$, the second largest retained block $b = c_2$ satisfies $b \geq h - 7 \geq 18$.

Put $v = N - b$. Each outside block has size at most b, so their degree is at least $Nv/2$. The preceding upper bound gives $v \leq 2hR/N + 2q/N < 2R + 2$, hence $v \leq 19 \leq b + 1$. If $b < h$, terminality gives $q \leq b + v - 1 \leq 2b$ and the first block has size $h \geq b + 1$. Even $q = 2t$ meets the two-block bounds $t + 1, t$; odd $q = 2t + 1$ meets $t + 2, t + 1$. If $b = h$, then $v \leq 19 \leq h - 1$, so $q \leq 2h - 2$ and the same size conditions hold.

Apply Lemma II.2.1. For odd q, check $d \neq 0$: with saturated target $K = p - 1$, $q \geq 2$ and $q < p$ give $p < d < 2p$. With $K \leq p - 2$ and $d < 2p$, the only exception is $d = p$, covered by Lemma II.2.7. □

II.6 Complete integer coverage for heights three through twenty-four

Lemma II.6.1 (Finite aggregate coverage)**.** *Let $3 \le h \le 24$, retain all original core conditions, and suppose $q > H$. If $b = c_2$ is the second largest original capacity, then*

$$q < L(h,b), \qquad L(h,b) = \begin{cases} 2b, & b = h, \\ 2b+1, & b = h-1, \\ 2b+2, & b \le h-2. \end{cases} \tag{II.6.1}$$

The exact finite integer exclusion below proves the assertion. Its verifier does not enumerate point sets or coefficient values.

Proof. Write the original nonzero capacities as $x_1 \ge x_2 \ge \cdots$, each at most b, with sum n. The Ferrers bound $B = n + \sum_j (j-1)x_j$ is minimized by filling capacities successively to b. Writing $n = tb + u$, $0 \le u < b$, gives

$$B \ge n + b\binom{t}{2} + tu = \frac{n^2 + bn + u(b-u)}{2b} \ge \frac{n(n+b)}{2b}.$$

Since $B < K \le nh$,

$$n \le n_{\mathrm{cap}}(h,b) = \min(14h-1, 2h^2-h-1, (2h-1)b-1). \tag{II.6.2}$$

The bound b applies to all original nonzero capacities, not merely a local maximum after omissions.

In the prefix put $c = c_j$, $z = c - s$, $w = \sum_{i=3}^{j-1} c_i$, and $R_0 = \sum_{i>j} c_i$. Then

$$1 \le c \le b \le h, \quad 0 \le z < c, \quad N = b + w + c - z, \quad w = 0 \text{ or } w \ge c, \quad n = N + z + R_0.$$

Set $n_{\max} = \min(14h-1, 2h^2-h-1)$. Since $d \ge (h+N)N/2$ and $d = K + q \le nh + N - 1$, $(h+N-2)N < 2hn_{\max}$. Define $N_{\max}$ as the largest integer in $[h+2, n_{\max}]$ satisfying this strict inequality. The left side is increasing, hence $N \le N_{\max}$. Also (II.5.1) gives

$$R_0 \ge R_{\mathrm{lo}} := \max\left(0, \left\lfloor \frac{N(N-h-2)}{2h} \right\rfloor + 1 - z\right). \tag{II.6.3}$$

The $+1$ coming from strictness must be retained.

Entirely omitted capacities are at most c, and their cost per slot is at least

$$\beta(a,c) = \begin{cases} a - (c-1)/2, & c \le a, \\ a(a+1)/(2c), & c \ge a. \end{cases}$$

Take this to be zero when $a = 0$. A full intermediate capacity $x \ge c$ costs at least $F_a(c)/c$ per slot. This is immediate for $c \le a+1$; otherwise use

$$\frac{F_a(x+1)}{x+1} - \frac{F_a(x)}{x} = \frac{x(x+1) - a(a+1)}{2x(x+1)} \ge 0.$$

Thus, with denominator $2c$ and $s = c - z$, put

$$\text{fixed} = 2c\{F_a(h) + F_a(b) + P_a(c,s)\} + 2F_a(c)w,$$

$$\text{beta} = \begin{cases} c(2a-c+1), & c \le a, \\ a(a+1), & c > a, \end{cases}$$

Then $2cC \geq \text{fixed} + \text{beta}\, R_0$.

Suppose $q \geq L(h,b)$. Put $q_{\min} = \max(h+1, L(h,b))$ and budget $= N + h - a - q_{\min} - 2$. Equation (II.4.2) forces

$$\text{fixed} + \text{beta}\, R_0 \leq 2c\, \text{budget}.$$

If fixed $> 2c\,$budget this is impossible. Otherwise an upper bound for R_0 is

$$R_{\text{hi}} = n_{\text{cap}}(h,b) - N - z,$$

and, if beta > 0, also $\lfloor (2c\,\text{budget} - \text{fixed})/\text{beta} \rfloor$; take the smaller bound. No division is made when beta $= 0$. Existence of an admissible R_0 therefore requires $R_{\text{lo}} \leq R_{\text{hi}}$.

The ancillary program `verify_finite.py` checks that this condition never holds on the complete finite domain

$$3 \leq h \leq 24, \quad 0 \leq a < h, \quad 1 \leq b \leq h, \quad 1 \leq c \leq b, \quad 0 \leq z < c,$$
$$\max(h+2, b+c-z, q_{\min}+1) \leq N \leq N_{\max}.$$

The only states excluded in advance are those proved impossible by $a + \binom{h-a}{2} > N_{\max} - 3$ or $0 < w < c$. All decisions use exact integers. For $h = 3, \ldots, 24$, the values of $N_{\max}$ are

$$8, 13, 19, 25, 33, 39, 44, 48, 53, 58, 63, 68, 72, 77, 82, 87, 92, 97, 101, 106, 111, 116.$$

The program reports, at every height, that each admissible interval is empty. Every actual configuration maps into this relaxed domain; no realizability assumption is imposed on unknown intermediate or omitted blocks. Thus this finite exclusion proves (II.6.1). □

To convert (II.6.1) to algebraic certificates, recall that a prefix has at least three blocks. If $b = h$, use two blocks for even $q \leq 2b - 2$, and the three-block balanced lemma for odd $q \leq 2b - 1$. If $b = h - 1$, use two blocks for even $q \leq 2b$, and the balanced lemma for odd $q \leq 2b - 1$. If $b \leq h - 2$, use two blocks for even $q \leq 2b$, and the asymmetric odd certificate for odd $q \leq 2b + 1$; its outside-slot condition follows from $N > q$. The check $d \neq 0$ for odd degrees is exactly that in Lemma II.5.1. The sole nonsaturated exception $d = p$ is covered by Lemma II.2.7. Every branch $q > H$ therefore has the required bound.

Part III
Saturated target at degree equal to the characteristic

III.1 Repairing the saturated degree-p case

It remains to treat $K = p - 1$ with globally minimal reachable degree $d = p$. The original core gives

$$h \geq 3, \quad n \geq h + 2, \quad n + h < p, \quad (n-1)h < p - 1 \leq nh. \tag{III.1.1}$$

Every capacity is at most h, and the zero-weight capacity is exactly h. All new configurations below are retained within these same capacities and have the original target $p - 1$. For the residual classification, suppose that no feasible retained configuration in the original capacities gives a certificate for this target. Any feasible repair producing a certificate excludes the original problem. This global failure assumption is necessary for an actual counterexample.

III.1.1 Swaps on the same positive support and reintroducing omitted classes

Two feasible configurations on the same positive support, differing only by exchanging unequal positive multiplicities r_i, r_j, cannot both fail at first order. Otherwise both satisfy (II.1.3); subtraction of their ith equations gives

$$0 = \frac{r_j - r_i}{u_i - u_j},$$

which is impossible since $0 < |r_i - r_j| < p$ and $u_i \neq u_j$. Both exchanged multiplicities must fit their respective original capacities. The coefficient values themselves are not exchanged.

Any degree-p configuration has $M - r_i \geq 2$ for every block: otherwise there is just one outside singleton and its degree is $r_i < M < p$. Thus it is automatically strictly terminal at target $p - 1$.

If such a configuration omits an original coefficient value and has at least two nonsingleton blocks, choose a donor block of size $r \geq 2$ while leaving another largest block $H \geq r$ unchanged. Transfer one donor slot to the omitted value. Total size stays fixed, the new degree is $p+r-1$, and the excess is r. A largest block still has size at least r, and at least r slots lie outside it. For even $r = 2t$, Lemma II.2.2 applies. For odd $r = 2t+1 \geq 3$, there are at least $2t+1 \geq t+2$ outside slots and the degree is nonzero modulo p, so the same lemma applies. With at most one nonsingleton block, Lemma II.2.8 already gives a certificate. Thus an unresolved configuration retains every original value.

III.1.2 Smoothing partially retained classes

Any degree-p configuration with largest block $H \leq h$ and N outside slots satisfies

$$N > h/2. \tag{III.1.2}$$

Otherwise $p \leq HN + N^2/2 \leq 5h^2/8$, whereas (III.1.1) gives $p > h(h+1)$.

Suppose a size-b block is partially retained and another has size $a \geq b$. If $a = b$, transfer one slot into the partial block: degree drops to $p - 1$ and volume applies. If $a = b + 1$, exchanging the multiplicities is feasible, giving a first-order certificate by the preceding subsection. If $a \geq b+2$, transfer one slot; the new degree is $p+a-b-1$ and excess $q' = a-b$ satisfies $2 \leq q' \leq h - 1$. The new largest size H' is at least $a - 1 \geq q'$ and at most the previous maximum, so $N' \geq N > h/2$. For even $q' = 2t$, $N' \geq t$; for odd $q' = 2t + 1$, $t + 2 \leq \lfloor h/2 \rfloor + 1 \leq N'$. The new degree is nonzero modulo p, and Lemma II.2.2 gives the original target.

Consequently each partially retained block in an unresolved configuration is strictly larger than every other retained block. There is at most one such block.

III.1.3 The precise residual domain and exclusion of two nonzero classes

There are at least two original nonzero classes, because their total $n > h$ while each capacity is at most h. With exactly two classes, both have size at least $n - h \geq 2$, so their mutual degree $D \geq 4$; with at least three classes, $D \geq 3$. The full original configuration thus has degree $nh + D > p$, since $p \leq nh + 1$, and cannot be the current degree-p configuration. Some block must be partial. It is the zero-weight block: otherwise the full zero block of size h would be larger, contradicting the preceding subsection.

Under the global failure assumption, every retained degree-p configuration therefore has the form

$$(a, c_1, \ldots, c_s), \qquad b = c_1 \geq \cdots \geq c_s \geq 1, \quad b < a < h, \tag{III.1.3}$$

with all nonzero classes full and of total n. Put $\delta = h - a$. Then

$$p = an + D, \quad D = \sum_{i<j} c_i c_j, \quad \delta n - h + 2 \le D \le \delta n + 1. \tag{III.1.4}$$

The lower bound is the integer form of $p > (n-1)h + 1$; the upper bound uses $p \le nh + 1$. Since $D \ge n(n-b)/2$, integrality gives $n - b \le 2\delta$: otherwise $D \ge \delta n + n/2 > \delta n + 1$, using $n \ge 5$. Hence

$$n - b \le 2\delta, \quad n \le h + \delta - 1, \quad n - b \ge \delta + 3, \quad \delta \ge 3, \quad n < 2h.$$

If $b = 1$, only the zero block can be nonsingleton, already covered by the source lemma. Suppose $b \ge 2$ and put $v = n - b$. If $v \le a - 1$, increase the zero block from a to $a + 2$ and decrease the size-b class to $b - 1$. This is feasible because $\delta \ge 3$. Total size increases by one and stays below $m < p$. The new degree and excess are

$$d' = p + n - a + b - 2, \qquad q' = n - a + b - 1 \le 2b - 2.$$

Also $q' \ge b + 4 \ge 6$. The two retained blocks $a + 2, b - 1$ meet the even or odd size bounds in Lemma II.2.1. Since $q' \le 2h - 4 < p$, the degree d' is nonzero modulo p. This again gives the original target.

Thus the only possible residual domain also satisfies

$$b \ge 2, \quad a \le v = n - b \le 2\delta, \quad \delta \ge \max(3, \lceil h/3 \rceil), \quad n < 2h. \tag{III.1.5}$$

If originally there were exactly two nonzero classes, then v is the second class size, so $v \le b < a$, contradicting $v \ge a$. The final residual domain therefore has at least three nonzero classes. This argument starts with two classes and supplies the required boundary case without assuming a general two-weight theorem.

III.2 Crossing at fixed total for heights at least sixteen

Lemma III.2.1 (Fixed-total crossing)**.** *In the residual domain* (III.1.3)–(III.1.5)*, if* $h \ge 16$*, a certificate for the original target exists.*

Proof. Add a zero slot, giving total $M_0 = n + a + 1 = m - \delta + 1 < p$ and degree $p + n$. Then make exactly $\delta - 1$ transfers, each deleting one slot from a positive nonzero class and adding one to the zero class. Total size stays fixed and the final zero size is h. There are enough nonzero slots since $\delta - 1 < n$. Throughout, the zero class is strictly larger than every nonzero class.

When the zero size is H and donor size is $r \ge 1$, the degree drops by $H - r + 1$, an integer from 1 to H. Let D' be the final degree among nonzero classes. Before deletion j, their total is $n - j$ and the donor size is at most b, so

$$D - D' \ge (\delta - 1)(n - b) - \binom{\delta - 1}{2}.$$

Putting $v = n - b$, the final total degree satisfies

$$\begin{aligned} d_{\text{end}} - p &\le h + v - \delta(h - b) + \binom{\delta - 1}{2} \\ &\le h + 1 - \binom{\delta + 1}{2} \le 0. \end{aligned}$$

The second inequality uses $h - b \geq \delta + 1$ and $v \leq 2\delta$. The last uses $\delta \geq 6$ for $h = 16$, and for $h \geq 17$, $\binom{\delta+1}{2} - h - 1 \geq (h^2 - 15h - 18)/18 > 0$.

The strictly decreasing path therefore first crosses from above p to at most p. Write d_-, d_+ for the degrees before and after, Δ for the drop, and $H \leq h - 1$ for the zero size before transfer. Then $1 \leq \Delta \leq H$. If $d_+ = p - 1$, use volume. If $d_+ \leq p - 2$, the preceding configuration has excess $q = d_- - p + 1$ with $2 \leq q \leq \Delta - 1 \leq H - 1$. Its outside size remains greater than $h/2$, since otherwise $d_- \leq 5h^2/8 < p$. Both parity conditions in Lemma II.2.2 hold, and $d_- = p - 1 + q \not\equiv 0 \pmod{p}$, giving a certificate.

If $d_+ = p$, a nonzero class is partial or omitted. A positive partial nonzero class is smaller than the zero class, so the smoothing argument applies. If every altered nonzero class has been emptied, an original value is omitted: reintroduce it when there are two nonsingleton blocks, or use Lemma II.2.8 otherwise. Automatic terminality at degree p was proved above. These cases exhaust the first crossing and always attain the original target. □

III.3 The remaining 524 small-height capacity certificates

It remains to consider $3 \leq h \leq 15$. Equations (III.1.3)–(III.1.5) give the complete finite integer domain

$$\begin{gathered} 3 \leq h \leq 15, \quad h + 2 \leq n < 2h, \quad n \leq 29, \quad s \geq 3, \\ c_1 \geq \cdots \geq c_s \geq 1, \quad b = c_1 \geq 2, \quad b < a < h, \quad \sum_i c_i = n, \\ \delta = h - a \geq \max(3, \lceil h/3 \rceil), \quad a \leq n - b \leq 2\delta, \\ p = an + \sum_{i<j} c_i c_j \text{ prime}, \quad (n-1)h < p - 1 \leq nh, \quad n + h < p. \end{gathered} \tag{III.3.1}$$

We also retain the redundant bound (III.1.4). This domain does not require the globally minimal degree to be p, so it contains the residual domain as a subset.

The ancillary program `verify_capacities.py` reconstructs this domain from all integer partitions of $n = 5, \ldots, 29$, enumerates h, a, checks primality, and computes the degree by integer products. The domain has exactly 524 elements. The file `capacity_witnesses.json` supplies a retained vector for each member.

Each record contains the original parameters, a retained vector in the original class order, and a certificate type. The program checks feasibility $0 \leq r_i \leq c_i$, total $M < p$, original target $k = p - 1$, excess $q = d - k$, and the hypotheses for its certificate:

1. `volume`: $d = p - 1$, in 461 records, using (II.1.2).
2. `order_two`: $q = 2$, at least two blocks, and a nonsingleton block, in 49 records, using the $(2, 1)$ case of Lemma II.2.1.
3. `terminal_3` or `terminal_4`: $q = 3$ or 4 with $q < M - H$, in 13 records, respectively twelve and one, using Lemma II.2.6.
4. `odd_large_block`: one record, with $h = 15$, $a = 4$, $(c_i) = (2, 2, 1^{18})$, $n = 22$, and $p = 317$. Retain $(8, 2, 2, 1^{15})$. Then $M = 27$, $d = 321$, $k = 316$, $q = 5$, largest size $8 \geq 5$, outside size $19 \geq 4$, and $d \not\equiv 0 \pmod{317}$. Apply Lemma II.2.2; no general fifth-order nonvanishing assertion is used.

h	Total	Volume	Order two	Order three/four	Odd large block
3–6	0	0	0	0	0
7	3	0	3	0	0
8	5	4	1	0	0
9	9	4	4	1	0
10	15	10	2	3	0
11	29	21	6	2	0
12	58	49	6	3	0
13	70	63	7	0	0
14	141	131	8	2	0
15	194	179	12	2	1
Total	524	461	49	13	1

The program uses bitset dynamic programming: after processing some capacities, bit d at total M records reachability of degree d. Adding a block of size r shifts the bitset by Mr. Degrees only increase, so bits above the target may be discarded. Exactly 461 domain members reach $p-1$. For the other 63, the known degree-p source configuration shows that the minimum reachable degree at least $p-1$ is exactly p. This classification is independently checked alongside the certificates.

The saved records are compared with the independently reconstructed domain and witnesses. These calculations use exact integers and make no tests of actual coefficient values or point sets. Each algebraic lemma applies uniformly to every set of distinct block values and every actual point set.

Part IV
Completion and exact verification

IV.1 Combining the proof and checking the boundaries

We complete the proof of Theorem I.1.1. If the assertion fails, choose a counterexample of minimal variable count in the same source field, across all set sizes and excesses. The core proof handles characteristic zero, the characteristic exclusion, the two-variable correction, and all initial boundaries. Prime-field reduction preserves every needed parameter and capacity. Section II.3 gives $n < 14h$. Here $h \geq 3$, $n \geq 6$, and the correction is zero, so the exact target is $K+1$, with $K = \min(nh, p-1)$.

Pad the original weights with h zeros and choose the globally minimal feasible degree $d = K+q$ from Section II.4, with largest size H. By (II.4.1), q is a nonnegative integer and $q < M - H < p$. The possibilities are:

1. $q = 0$: ordinary volume at degree $K < p$ gives at least $K+1$ values.
2. $q = 1$ and $K \leq p-2$: degree $d = K+1 < p$ gives at least $K+2$ values by volume.
3. $2 \leq q \leq H$: terminality gives more than q outside slots, meeting both parity conditions in Lemma II.2.2. For odd q with $d \equiv 0 \pmod{p}$, the bound $d < 2p$ forces $d = p$. This cannot occur for the saturated target; for the nonsaturated target, Lemma II.2.7 gives at least $p-1 \geq K+1$ values.
4. $q > H$: use the prefix lemma and total-size bound. For $h \geq 25$ apply Section II.5; for $3 \leq h \leq 24$ apply the complete integer coverage and algebraic cases of Section II.6. The same nonsaturated degree-p exception is rescued by the stronger bound.
5. $q = 1$ and $K = p-1$: Section III.1 either gives a certificate or reduces to the precise

residual domain. For $h \geq 16$ use fixed-total crossing; for $3 \leq h \leq 15$ use all 524 certificates in Section III.3.

In every branch the retained slots fit the original capacities, and fixing omitted slots embeds a translate of their actual support into the original support. No point is reused and no equal-valued class is split. The five branches exhaust all q and $h \geq 3$, each giving at least $K + 1$ values, contrary to the counterexample. This proves the theorem.

IV.2 Reproducible exact verification

The ancillary program `verify_finite.py` reconstructs and checks the 72 uniform polynomial identities, all 3,766 endpoint certificates (covering 39,541 integer sumset sizes), the characteristic-eleven polynomial identities, and the relaxed aggregate domain for heights three through twenty-four. The aggregate search visits 13,978,262 states and leaves no admissible interval. Its largest bounds are $n_{\max} = 335$ and $N_{\max} = 116$.

The program `verify_capacities.py` independently reconstructs the 524-member small-height domain and supplies feasible witnesses. It compares its output with `capacity_witnesses.json` and checks the degree, original target and lemma hypotheses for every witness. In this domain $p \leq 15 \cdot 29 + 1 = 436$. The file `verify_all.py` runs both checks. Python 3 and SymPy are required for the polynomial expansions; the capacity check uses only the Python standard library.

These programs verify the explicitly finite arithmetic and polynomial claims. The unbounded reductions, matching identities, capacity arguments, and their hypotheses are proved in the text. Every search bound is established before its finite verification is invoked.

A Proof of the minimal-core lemma

We prove Lemma I.2.1, supplying the necessary elementary lemmas and the precise six-point configuration left by a five-variable minimal counterexample. No general small-characteristic or two-weight theorem is assumed without proof.

A.1 Published inputs with their exact hypotheses

We use the following statements of Sun–Zhao, version seven [1], valid over arbitrary fields for nonzero weights:

1. Theorem 1.1: if $p(F) \geq n(3n-5)/2$, the bound (I.1.1) holds, including δ for two opposite weights.
2. Theorem 1.2: if $p(F) \geq (n-1)^2$ and every finite set A_i has at least $2n-2$ elements, the weighted sumset from distinct choices has at least
$$\min\left(p(F) - \delta, \sum_i |A_i| - n^2 + 1\right)$$
elements. Here we use only $A_i = A$ for every i.
3. Equation (1.8): for a common set of size m,
$$|S| \geq \min\left(p(F) - \binom{n}{2}, n(m-n) + 1\right).$$

These thresholds include equality. They settle characteristic zero and impose (I.2.3) and (I.2.4) on a positive-characteristic counterexample. In particular, the second strict failure condition $p < nh + \binom{n}{2} + 1$ is written integrally as $p \leq nh + \binom{n}{2}$.

For one specified six-point configuration we also use Fehér–Nagy, version four [2, Theorem 4.2]. For r nonzero weights and a common N-point set, with distinct-weight multiplicities $k_1, \dots, k_t$, put

$$d = r(N-r) + \sum_{i<j} k_i k_j.$$

If $p(F) > d$, the sumset has at least $d+1$ elements. We use this only for $r = 4$, $N = 6$, multiplicities $(2,1,1)$, hence $d = 13$, and $p = 17$ or 19. No use is made when $p \le d$.

A.2 Grid interpolation and large sets

If $P \in F[x_1, \dots, x_r]$ has total degree at most $\sum_i t_i$ and $|A_i| = t_i + 1$, then

$$\Big[x_1^{t_1} \cdots x_r^{t_r}\Big]\, P = \sum_{(z_i) \in \prod_i A_i} \frac{P(z_1, \dots, z_r)}{\prod_i \prod_{b \in A_i \setminus \{z_i\}} (z_i - b)}. \tag{A.1}$$

This follows by Lagrange interpolation in each variable: the corresponding univariate functional kills degrees below t_i and takes $x_i^{t_i}$ to one. Every other monomial of total degree at most $\sum_i t_i$ has some exponent below t_i and is killed by the product functional. Distinctness within each set makes all denominators nonzero.

Put $\Delta_r(x) = \prod_{i<j}(x_j - x_i)$ and $H = \sum_i b_i x_i$. If

$$H^k \Delta_r \neq 0 \quad \text{in } F[x_1, \dots, x_r]/(x_1^M, \dots, x_r^M), \tag{A.2}$$

then the restricted weighted sumset on every M-point set has at least $k+1$ elements. Otherwise choose a monic degree-k polynomial Q vanishing on the sumset. The polynomial $Q(H)\Delta_r$ vanishes on the full grid, but its top homogeneous part has a nonzero coefficient with every exponent below M. Applying (A.1) to the corresponding subgrid is a contradiction. No extra bound $k \le r(M-r)$ is required.

Lemma A.1 (Large-set bound)**.** *Over any field of characteristic p, if $|A| = p$, $1 \le r \le p$, and $\sum_i b_i \neq 0$, then $|S(b; A)| \ge p$. If $m > p$ and $m > n$, then $|S(a; A)| \ge p$ for all nonzero weights.*

Proof. In $R = F[x_1, \dots, x_r]/(x_1^p, \dots, x_r^p)$, the derivation $D = \sum_i \partial_{x_i}$ is well defined, since $D(x_i^p) = 0$. We have $D\Delta_r = 0$ and $DH = \sigma = \sum_i b_i$, so

$$D^{p-1}(H^{p-1}\Delta_r) = (p-1)!\sigma^{p-1}\Delta_r \neq 0.$$

The monomial exponents of Δ_r permute $0, \dots, r-1 < p$, so it is nonzero in R. The criterion (A.2) proves the first assertion.

For the second, choose $q_0 = \min(n, p)$ weights and, if their sum is zero, delete one. The remaining $r = q_0$ or $q_0 - 1 \ge 1$ weights have nonzero sum. Fix the other $n - r$ slots while leaving at least p points: for $n \le p$, $n - r \le 1 \le m - p$; for $n > p$, $n - r \le n - p + 1 \le m - p$. Apply the first assertion to any p unused points. A translate of these sums lies in the original S. □

A.3 Initial ranges and zero-padding translations

For $n = 1$, $|S| = m$; for $m = n$, S is nonempty. Assume $m > n$. Sun–Zhao Theorem 1.1 covers $n = 2$. For $n = 3$, use that theorem for $p \ge 7$ and Theorem 1.2 for $p = 5, m \ge 4$. For $p = 2$, fix two coordinates and vary the third to obtain two values. For $p = 3$, some pair of nonzero weights has nonzero sum: otherwise all three pair sums vanish, forcing a weight to

be zero in odd characteristic. Fix the third coordinate and apply Theorem 1.1 to that pair to get three values. This is also the argument following Remark 1.1 of [1].

Thus a minimal counterexample has $n \geq 4$ and $m > n$. The large-set bound gives $m \leq p$, and the characteristic exclusion gives $n \leq p-2$, hence $p \geq 7$. Its correction term is zero.

Pad the n weights by $h = m - n$ zeros. The original sumset is exactly the permutation sumset of these m weights on A. Subtracting the same c from every padded weight translates this set by $-c\sum_{x\in A} x$.

If a nonzero weight c has multiplicity $r > h$, this translation creates exactly r zeros. Delete them to leave $n' = m - r < n$ nonzero weights on A, with new excess r. Of these weights, h equal $-c$, and

$$n'r - nh = (r-h)(n-r) \geq 0. \tag{A.3}$$

Since $n' < n \leq p-2$, the smaller dimension is not the excluded $p-1$. If its correction is zero, minimality and (A.3) immediately contradict the counterexample.

If its correction is one, $n' = 2$ with opposite weights. Since $p \geq 7$, $h = 2$ would make both weights equal to $-c$, not opposite. Thus $h = 1$ and $n - r = 1$. Minimality still gives

$$|S| \geq \min(p-1, 2n-1) \geq n+1 = nh+1,$$

where $p - 1 \geq n + 1$ and $n \geq 4$, again a contradiction. Consequently

$$\text{every nonzero-weight multiplicity satisfies } r_i \leq h. \tag{A.4}$$

This translation changes the excess, explaining why minimality must cover all excesses.

Fix one position and its chosen point. The remaining $n - 1 \geq 3$ weights run on $m - 1$ points with excess h. Their characteristic exclusion holds and correction is zero, so minimality gives

$$s \geq \min(p, (n-1)h + 1).$$

A counterexample requires $s < p$, hence

$$K_0 = (n-1)h + 1 \leq s < p, \qquad p \geq (n-1)h + 2. \tag{A.5}$$

If $h \geq n-1$, then $p \geq (n-1)^2 + 2$ and $m \geq 2n-1$. Sun–Zhao Theorem 1.2 gives the required bound, a contradiction. Thus $h \leq n-2$.

A.4 Permutation blocks with distinct weights

Over a field of characteristic $p \geq 5$, let $u_1, \ldots, u_b$ and $x_1, \ldots, x_b$ each be pairwise distinct. Their permutation inner products have at least

$$\min\left(p, \binom{b}{2} + 1\right) \tag{A.6}$$

values. One weight may be zero. We give the algebraic proof, including the saturated case.

For r variables put $D = \binom{r}{2}$, $J_r = \prod_{j=0}^{r-1} j!$, $V(z) = \prod_{i<j}(z_j - z_i)$, and $L_\pi = \sum_i u_i x_{\pi(i)}$. Alternating expansion gives the integer identity

$$J_r \sum_\pi \operatorname{sgn}(\pi) L_\pi^D = D! V(u) V(x). \tag{A.7}$$

Alternating polynomials in x of degree below D vanish. At degree D, only exponent permutations of $0, 1, \ldots, r-1$ survive, with coefficient $D!/J_r$; two determinant expansions prove the identity.

If $p > \binom{b}{2}$, take $r = b$ in (A.7) and replace L_π^D by a monic degree-D polynomial $Q(L_\pi)$. Lower-degree terms alternate to zero. If Q vanished on all permutation values, the left side would vanish and the right side would not, a contradiction. The case $b = 1$ is immediate.

If $p \le \binom{b}{2}$, take the smallest $r \le b$ with $\binom{r}{2} \ge p$. Then $4 \le r < p$, and

$$1 \le k = D - p + 1 \le r - 1, \qquad D - k = p - 1.$$

Indeed, $r = 4$ gives $r < p$ since $p \ge 5$. For $r \ge 5$, $\binom{r-1}{2} \ge r$, while minimality makes this binomial coefficient smaller than p.

Differentiate the rational form of (A.7) k times in u_i, canceling first in characteristic zero, to obtain the integer identity

$$J_r \sum_\pi \operatorname{sgn}(\pi) x_{\pi(i)}^k L_\pi^{D-k} = (D-k)!\, \partial_i^k V(u) V(x).$$

The operator $\sum_i u_i^k \partial_i^k$ preserves degree and alternation. Comparing coefficients of $\prod_i u_i^{i-1}$ gives

$$\sum_i u_i^k \partial_i^k V(u) = \left(\sum_{j=0}^{r-1} (j)_k \right) V(u) = k! \binom{r}{k+1} V(u).$$

Thus, for a monic degree-$(p-1)$ polynomial Q,

$$J_r \sum_\pi \operatorname{sgn}(\pi) \left(\sum_i (u_i x_{\pi(i)})^k \right) Q(L_\pi) = (p-1)!k! \binom{r}{k+1} V(u) V(x). \tag{A.8}$$

The substituted lower terms have total x-degree below D, so alternate to zero. All cancellations take place over $\mathbb{Q}$ before the cleared integer identity is reduced modulo p. No zero factorial is canceled in characteristic p.

The right side is nonzero since $k < r < p$ and both lists are distinct. At most $p - 1$ permutation values would admit a vanishing monic degree-$(p-1)$ polynomial Q, a contradiction. Fix the other $b - r$ pairings to embed a translate of these at least p values into the original sumset. This proves (A.6), including $b = 2, 3$ in large characteristic, without using an extension that fails for $p = 3, b = 3$.

If $h = 1$, (A.4) makes the original weights and the added zero all distinct. Apply (A.6) to $m = n + 1 \ge 5$ to obtain $s \ge \min(p, m) = m = nh + 1$, a contradiction. Thus $h \ge 2$.

If $n = 4$, then $2 \le h \le n - 2$ forces $h = 2, m = 6$. Equation (A.5) gives $p \ge 8$, so $p \ge 11 > 9 = (n-1)^2$. Sun–Zhao Theorem 1.2 excludes this case. Hence $n \ge 5$.

A.5 Multiplicity blocks and the five-variable residual case

Let the nonzero-weight multiplicities be $r_1, \dots, r_t \le h$, and put $t_j = \#\{i : r_i \ge j\}$ for $j = 1, \dots, h$. Partition the zero-padded weights into h blocks: block j contains a zero and one copy of every weight with $r_i \ge j$. Its $1 + t_j$ weights are distinct; total size is m. Partition A into matching point-block sizes and permute within blocks independently. The sum of their value sets lies in S.

By (A.6) and (I.4.1), valid over any characteristic-p field,

$$s \ge \min(p, B + 1), \qquad B = \sum_{j=1}^{h} \binom{t_j + 1}{2} = n + \sum_{i<k} \min(r_i, r_k). \tag{A.9}$$

If one block already yields p values, the result follows. Otherwise iterated Cauchy–Davenport adds the block sizes minus one, giving this bound. A counterexample forces

$$B < \min(nh, p - 1).$$

Since $\sum_j t_j = n$, Cauchy–Schwarz gives $B \geq (n^2/h + n)/2$, hence

$$n < 2h^2 - h. \tag{A.10}$$

For $n = 5$, we have $2 \leq h \leq 3$. If $h = 3$, (A.5) gives $p \geq 14$, whereas (I.2.4) gives $p < 16$, with no prime between. Thus $h = 2, m = 7$, and (I.2.3) and (A.5) give

$$p \in \{11, 13, 17, 19\}.$$

Each nonzero multiplicity is at most two. If there are $e = 0, 1, 2$ double classes, the quantity B in (A.9) is $15, 11, 9$, respectively. Since $B < nh = 10$, only multiplicities $(2, 2, 1)$ remain. Write the weights as r, r, s, s, t, with three distinct nonzero values. Fix one r position. On the remaining six points the weights are r, s, s, t; padding two zeros gives four distinct block values of multiplicities $(2, 2, 1, 1)$. A translate of their value set lies in S.

For $p = 17$ or 19, the cited Fehér–Nagy theorem has $d = 13 < p$ and gives at least 14 values, exceeding the required 11. For $p = 11$ or 13, the six-point lemma in Appendix B gives at least 11. Hence $n = 5$ is impossible and $n \geq 6$. By (A.10), $h = 2$ would force $n < 6$, so $h \geq 3$.

Finally, (A.5) gives

$$p - m \geq (n - 1)h + 2 - (n + h) = (n - 2)(h - 1) \geq 8.$$

Combining these bounds proves every assertion of Lemma I.2.1. In particular $K_0 \geq 16$ and $p > K_0$, so $p \geq 17$. Equal weights are impossible because their multiplicity n exceeds $h \leq n-2$. The argument excludes precisely the configurations left by a minimal counterexample; it does not assume a general five-variable theorem.

B Six-point identities for the five-variable residual case

Lemma B.1 (Six points and four blocks)**.** *In any field of characteristic* 11 *or* 13*, six distinct points and four distinct weight values of multiplicities* $(2, 2, 1, 1)$ *produce at least* 11 *permutation sums. One weight value may be zero.*

The proof uses polynomial identities, and therefore applies to arbitrary extension fields and infinite fields, rather than just prime-field parameter points.

B.1 Published identities and derivatives that admit reduction

Write the four block values as $\mu = (u, v, w, z)$ and their sizes as $(k_1, k_2, k_3, k_4) = (2, 2, 1, 1)$. The cross-block pair count is $d = 13$. Put

$$P(\mu) = (u - v)^4(u - w)^2(u - z)^2(v - w)^2(v - z)^2(w - z), \tag{B.1}$$

and

$$D = \prod_{j=1}^{5} j! = 34560 = 2^8 \cdot 3^3 \cdot 5.$$

For initially independent point variables $X_1, \ldots, X_6$, let $I = (I_1, I_2, I_3, I_4)$ range over ordered partitions of sizes $(2, 2, 1, 1)$, and define

$$T_j(I) = \sum_{i \in I_j} X_i, \qquad L_I(\mu) = \sum_j \mu_j T_j(I),$$

$$E_I(X) = \prod_{r<s} \prod_{i\in I_r,\ j\in I_s} (X_i - X_j).$$

Both point and weight differences use the direction "earlier block minus later block." The degree identity of Fehér–Nagy [2, equations (4.3), (4.5)–(4.7)], version four, becomes

$$\sum_I \frac{L_I(\mu)^{13}}{E_I(X)} = \frac{13!}{D} P(\mu), \tag{B.2}$$

$$\sum_I \frac{L_I(\mu)^j}{E_I(X)} = 0 \quad (0 \le j < 13). \tag{B.3}$$

First regard these as characteristic-zero rational-function identities. The denominator from cross-block position differences is D: the two within-block position differences in size-two blocks are both one, so removing them from the full product leaves $\prod_{j=1}^{5} j!$.

To justify the independent variables and lower-degree identities, use the source's deformation with 13 independent parameters $b_1, \ldots, b_{13}$, replacing L_I^{13} by $\prod_{j=1}^{13}(L_I - b_j)$. Comparing the specific squarefree coefficient of $b_1 \cdots b_{13-j}$ gives (B.3) without division by a binomial coefficient. If the source fixes the last block value at zero, substitute $\mu_i - z$ for the other values and $b_j - z\sum_{a=1}^{6} X_a$ for each b_j. This gives the identity with all four block values independent; its right side depends only on differences and is unchanged.

For a multi-index α of total order $q = |\alpha|$, put $k = 13 - q$. Differentiate (B.2) ordinarily in μ and cancel $13!/k!$ in characteristic zero to obtain

$$\sum_I \frac{T(I)^\alpha L_I(\mu)^k}{E_I(X)} = \frac{k!}{D}\, \partial^\alpha P(\mu). \tag{B.4}$$

For $0 \le j < k$, differentiate the degree-$(j+q)$ instance of (B.3) and cancel its falling factorial in characteristic zero. This gives

$$\sum_I \frac{T(I)^\alpha L_I(\mu)^j}{E_I(X)} = 0.$$

Each E_I divides the six-variable Vandermonde up to sign. Multiplication by that common Vandermonde, and by D if necessary, gives integer polynomial identities. Their validity over $\mathbb{Q}$ means every integer coefficient is zero, so they reduce to arbitrary characteristic. In particular, no vanishing $13!$ is canceled in characteristic 11 or 13.

Now specialize the X_i to any six distinct field elements. If $p \nmid D$ and $k < p$, every monic degree-k polynomial Q satisfies

$$\sum_I \frac{T(I)^\alpha Q(L_I(\mu))}{E_I(X)} = \frac{k!}{D}\, \partial^\alpha P(\mu). \tag{B.5}$$

If there were at most k permutation values, choose Q vanishing on all of them. Since $k!D \neq 0$, any nonzero order-q derivative gives at least $k+1$ values. Thus an order-two derivative suffices in characteristic 13, and an order-three derivative in characteristic 11.

B.2 Second derivatives in characteristic thirteen

Distinct block values give $P(\mu) \neq 0$. If all second derivatives P_{ij} vanished, Euler's formula applied to each degree-twelve polynomial P_j would give

$$\sum_i \mu_i P_{ij} = 12 P_j,$$

so all first derivatives would vanish too. For $i \neq j$, however, direct quotient differentiation gives

$$\frac{P_{ij}}{P} - \frac{P_i P_j}{P^2} = \frac{k_i k_j}{(\mu_i - \mu_j)^2}. \tag{B.6}$$

The numerator on the right is $1, 2$, or 4, nonzero in characteristic 13, and its denominator is nonzero. This contradiction shows that some second derivative is nonzero. With $q = 2, k = 11$, (B.5) gives at least 12 values, sufficient for the required 11.

B.3 A complete polynomial certificate in characteristic eleven

Put $h_0 = z - v \neq 0$, $a = (u - v)/h_0$, and $b = (w - v)/h_0$. Then

$$a, b, a - 1, b - 1, a - b \neq 0,$$

$$P(v + h_0 a, v, v + h_0 b, v + h_0) = h_0^{13} f(a, b), \qquad f = a^4 b^2 (a - b)^2 (a - 1)^2 (b - 1).$$

Holding v, h_0 fixed, for $i + j = 3$ we have

$$(\partial_u^i \partial_w^j P)(u, v, w, z) = h_0^{10} (\partial_a^i \partial_b^j f)(a, b).$$

It suffices to show that the four ordinary third derivatives do not vanish simultaneously. All following coefficients lie in $\mathbb{F}_{11}$. Their exact factorizations are

$$\begin{aligned} f_{aaa} &= ab^2(b-1)g_0, & f_{aab} &= a^2 b g_1, \\ f_{abb} &= a^3(a-1)g_2, & f_{bbb} &= a^4(a-1)^2 g_3, \end{aligned} \tag{B.7}$$

where

$$\begin{aligned} g_0 &= -5a^4 - 2a^3 b - 2a^3 - a^2 b^2 - 4a^2 b - a^2 + ab^2 + ab + 2b^2, \\ g_1 &= 3a^4 b - 2a^4 + 5a^3 b^2 + 3a^3 - 4a^2 b^3 - 3a^2 b^2 \\ &\quad + 5a^2 b - 5a^2 - 2ab^3 - ab + 5b^3 - 4b^2, \\ g_2 &= 4a^3 b - 5a^3 - 3a^2 b^2 + 4a^2 b + a^2 - ab^3 \\ &\quad + 4ab^2 - 5ab - 3b^3 + 4b^2, \\ g_3 &= -5a^2 - 4ab + a + 5b^2 - 2b. \end{aligned}$$

Define

$$\begin{aligned} H_0 &= -3ab - 4a + b^2 + 4b - 1, \\ H_1 &= 3a + 4b - 2, \\ H_2 &= 5a^2 + 5ab + 4a - 3b + 4, \\ H_3 &= -4a^2 b - 5ab^2 - 5ab - 4a - 4b^2. \end{aligned}$$

Termwise expansion gives the Bézout identity

$$H_0 g_0 + H_1 g_1 + H_2 g_2 + H_3 g_3 = b^2(a - b) \quad \text{in } \mathbb{F}_{11}[a, b]. \tag{B.8}$$

Equations (B.7) and (B.8) can be expanded by hand and are also checked coefficientwise by `verify_finite.py`. These are polynomial identities, rather than sampling assertions on finite-field points.

All factors removed from the derivatives are nonzero on this open set. If the four third derivatives vanished, all g_i would vanish; (B.8) would then give $b^2(a - b) = 0$, a contradiction. Some third derivative is therefore nonzero. Set $q = 3, k = 10$ in (B.5) to obtain at least 11 values. This proves the six-point lemma.

C A direct unweighted restricted-sum proof in the core range

Lemma C.1 (All weights equal to one)**.** *Let F have prime characteristic p, let $A \subseteq F$ have size m, and suppose $1 \le n \le m < p$. Then*

$$|\{x_1 + \cdots + x_n : x_i \in A \text{ pairwise distinct}\}| \ge \min(p, n(m-n)+1).$$

Proof. Put $h = m - n$ and $k = \min(p-1, nh)$, so $0 \le k < p$. Write uniquely $k = nq + r$, $0 \le r < n$. Choose

$$t_i = i - 1 + q + \begin{cases} 0, & i \le n - r, \\ 1, & i > n - r. \end{cases}$$

These integers are strictly increasing and satisfy $0 \le t_1 < \cdots < t_n \le m - 1 < p$. The final upper bound follows from $k \le nh$: if $r > 0$, $q + 1 \le h$; if $r = 0$, $q \le h$. Also

$$\sum_i t_i = \binom{n}{2} + k.$$

Let $L = x_1 + \cdots + x_n$ and $\Delta_n = \prod_{i<j}(x_j - x_i)$. Determinant expansion gives

$$\left[x_1^{t_1} \cdots x_n^{t_n}\right] L^k \Delta_n = k! \det\left(\frac{1}{(t_i - j + 1)!}\right)_{i,j=1}^{n} = \frac{k! \prod_{i<j}(t_j - t_i)}{\prod_i t_i!}. \tag{C.1}$$

A matrix entry involving a negative factorial is interpreted as zero. For the second equality, multiply row i by $t_i!$; its entries become the monic falling-factorial polynomials $(t_i)_{j-1}$. Column operations give the Vandermonde determinant. Equivalently, compute the integer coefficient over $\mathbb{Q}$ before reducing to F.

All factorial arguments are below p, and all differences $t_j - t_i$ lie in $1, \ldots, p-1$. Thus (C.1) is nonzero in F. The truncated-grid criterion in Appendix A gives at least $k+1 = \min(p, nh+1)$ restricted sums. □

The main text uses this lemma only after establishing $m < p$. No assertion about arbitrary large sets or a general two-weight theorem is needed here.

References

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